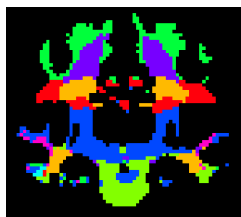


WhiFuN Manual



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Introduction

We present the White Matter Functional Networks (WhiFuN) Toolbox for automated preprocessing of WM and GM fMRI, robust construction of WM and GM Functional Networks (FN), computation of WM-FC and GM-FC, and FC analysis modules. WhiFuN is based on SPM12 preprocessing and contains statistical tools for group-level analyses. WhiFuN provides an intuitive graphical user interface allowing users to execute all steps from preprocessing to final group level analyses and does not require prior knowledge of computer programming.

WhiFuN was developed using MATLAB (version R2022b) under the Windows environment. The WhiFuN GUI was developed using MATLAB App Designer. In addition to custom-written code, several functions from SPM12 (<https://www.fil.ion.ucl.ac.uk/spm/>) were used for preprocessing and quality control. WhiFuN saves quality control plots to assess the quality of preprocessing steps. Users may select the data corresponding to subjects (using the *uigetfile_n_dir* function developed by Tiago (Tiago, 2024)) for which the preprocessing was not done as desired and discard them from further analysis. Visualization of the WM and GM-FNs was performed in conjunction with the BrainNet viewer toolbox (Xia et al., 2013).

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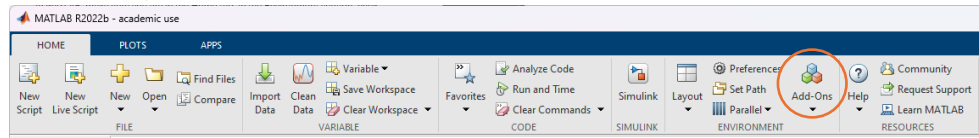
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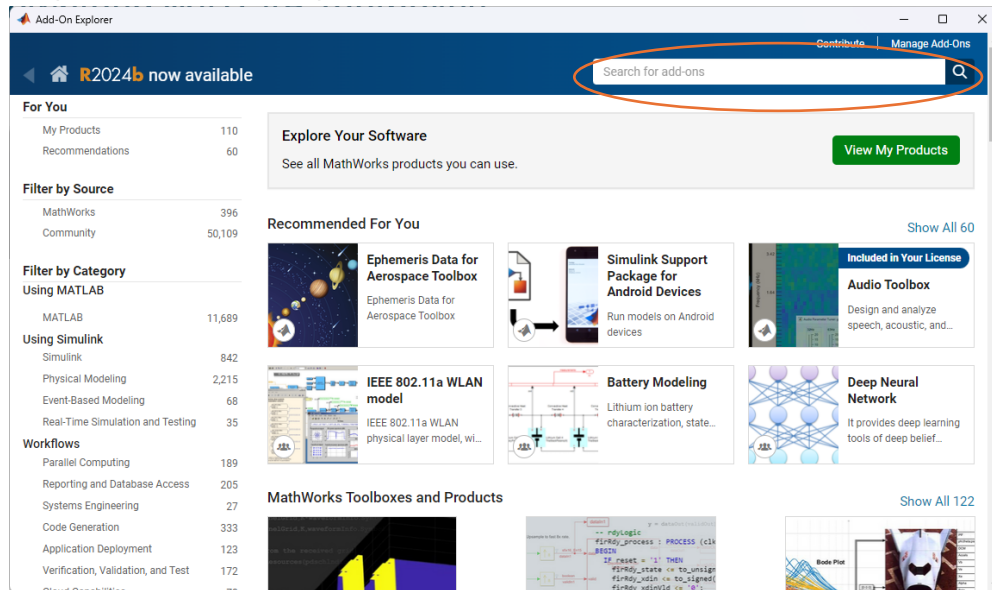
1. Getting started

1.1 Download MATLAB Toolboxes

- i) Run MATLAB
- ii) Make sure you are in the Home Tab. In the Environment section click on Add-Ons.



It opens the Add-Ons explorer.



- iii) From the search bar, type the names of the following toolboxes and install them in MATLAB.

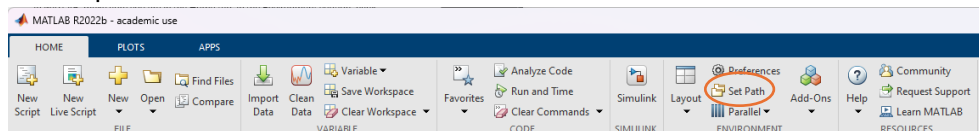
- 1) Image processing Toolbox
- 2) Signal Processing Toolbox
- 3) Statistics and Machine learning toolbox
- 4) Bioinformatics Toolbox

1.2 Download SPM 12 and Add path to MATLAB

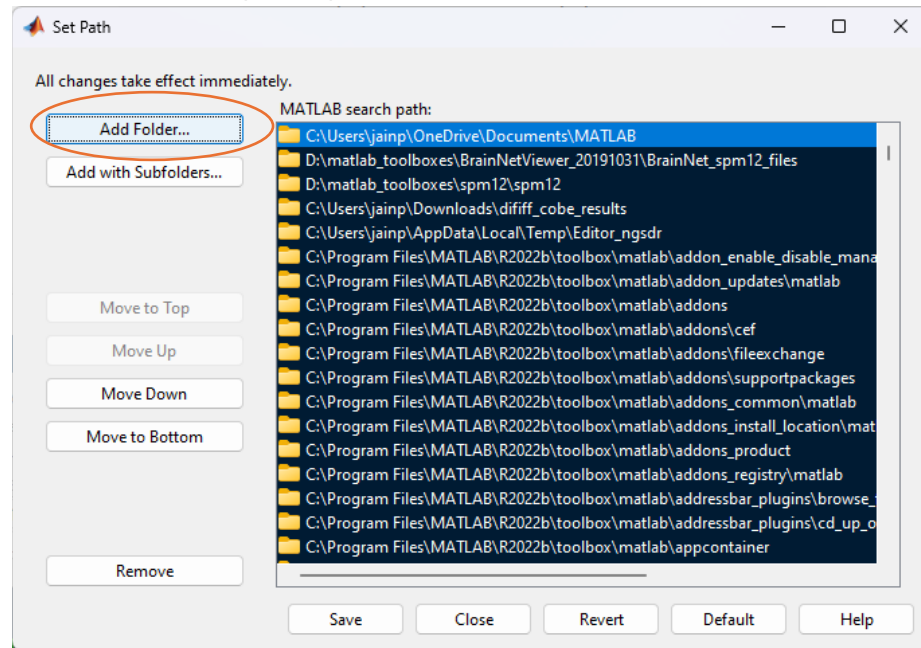
- i) Download SPM12 toolbox from (<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>).
- ii) Add path of SPM toolbox by

- 1) By MATLAB Set path (**Recommended**)

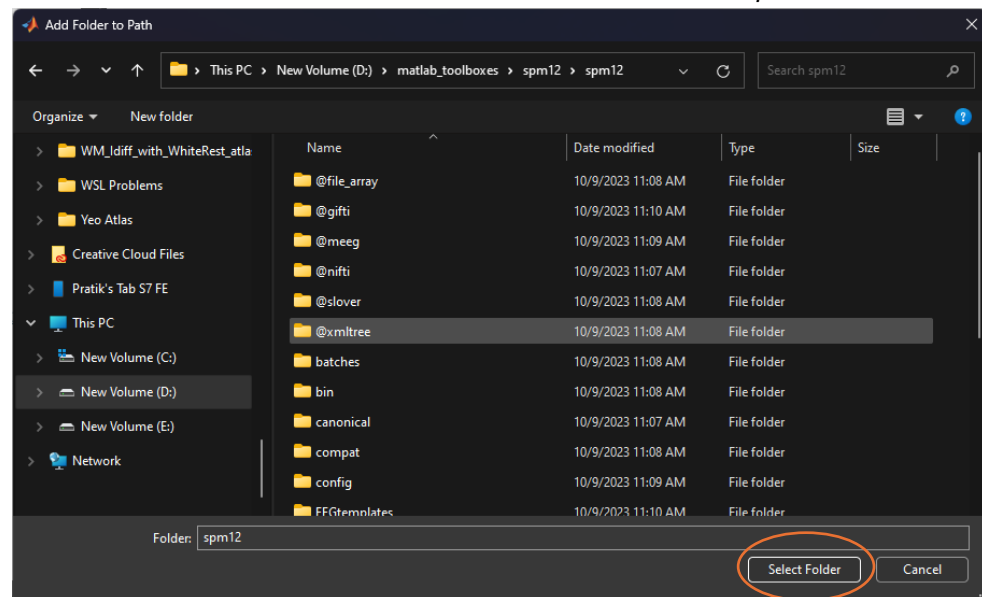
In MATLAB, make sure you are in the Home tab. In the Environment section, click on Set path.



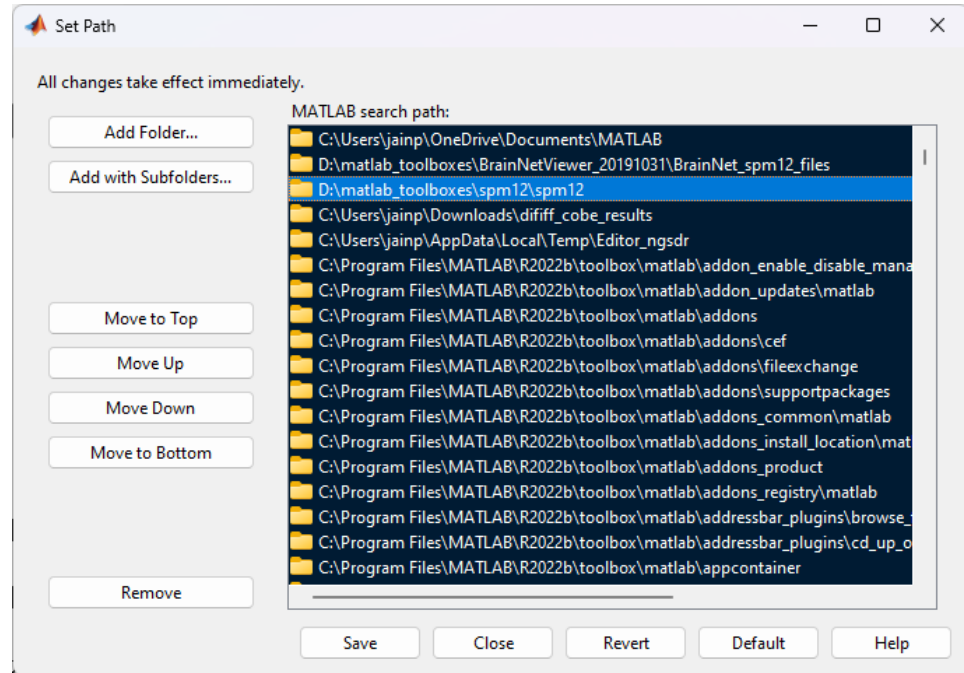
Set Path window opens up.



Click on Add folder and Select the SPM folder that has the *spm.m* code



You will see the folder added to the path. You can save and close the set path window.



or

2) By MATLAB Command (**Will have to do every time MATLAB is restarted**)

In MATLAB command window type

```
addpath('X:\ ... \spm12\spm12')
```

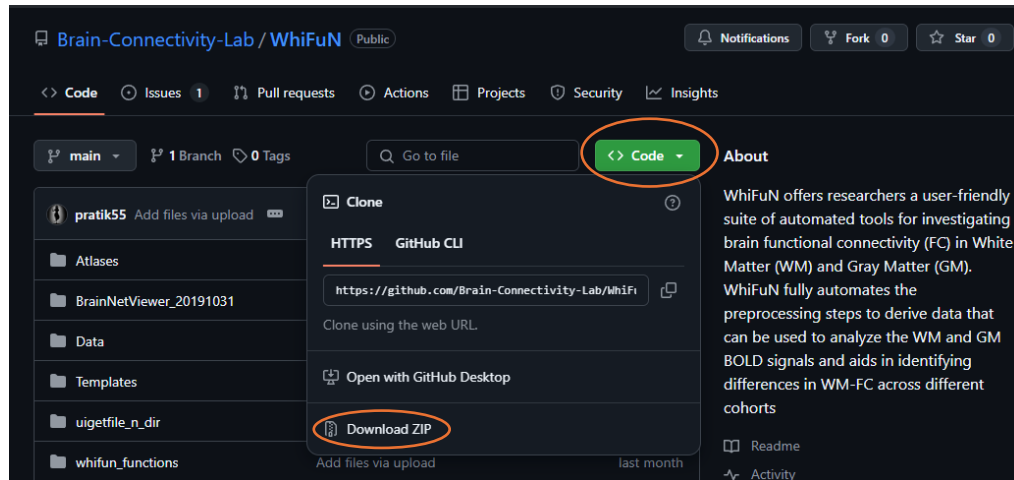
where, 'X:\ ... \spm12\spm12' is the path of the SPM 12 toolbox on your machine.

Eg.

```
addpath('D:\matlab_toolboxes\spm12\spm12')
```

1.3 Download WhiFuN and Add path to MATLAB

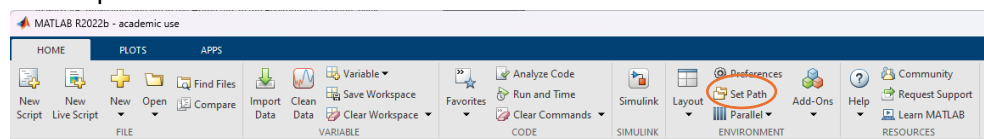
- i) Download WhiFuN from <https://github.com/Brain-Connectivity-Lab/WhiFuN>
(click on the green code button and, then download zip)



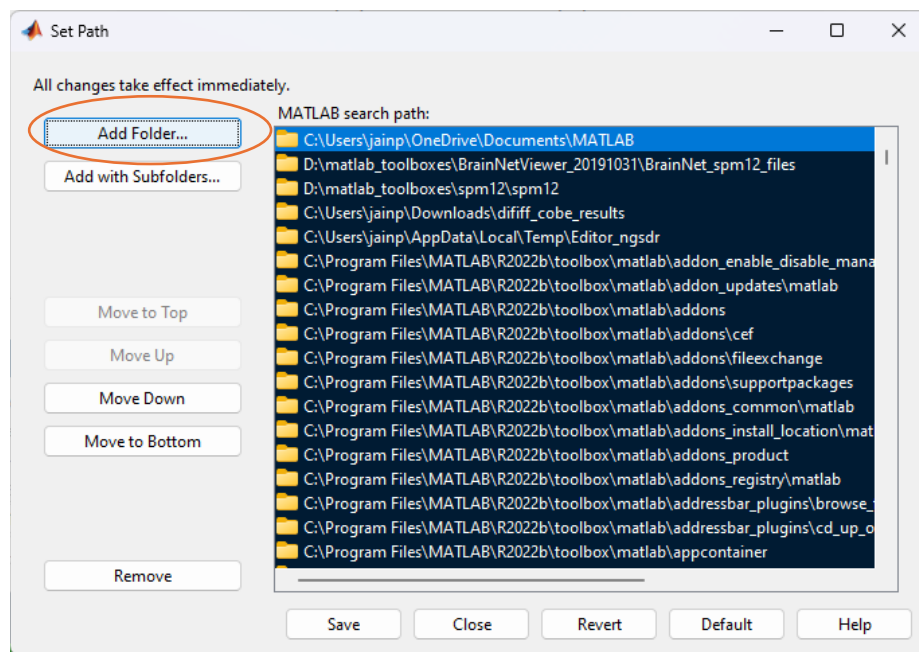
ii) Unzip the contents and add path to MATLAB by

iii) By MATLAB Set path (**Recommended**)

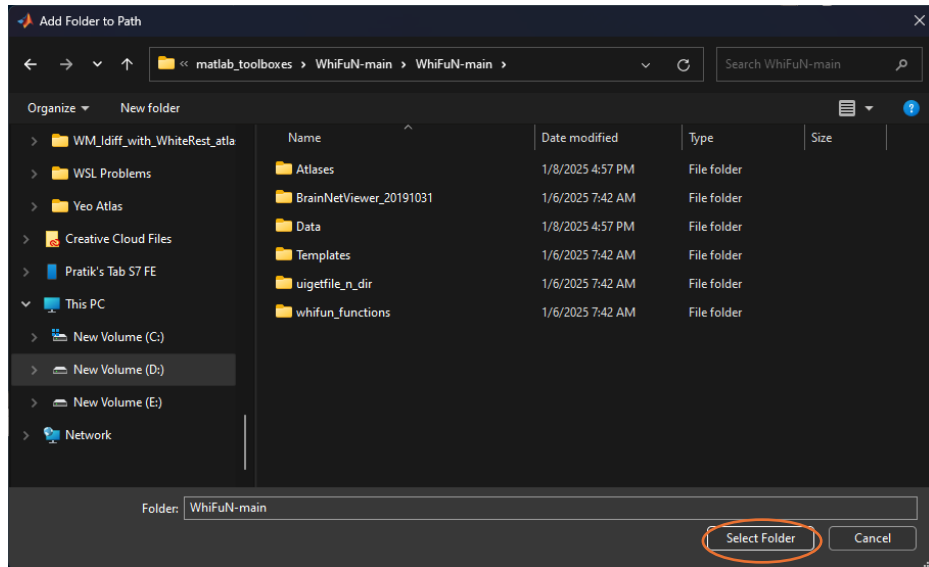
In MATLAB, make sure you are in the Home tab. In the Environment section, click on Set path.



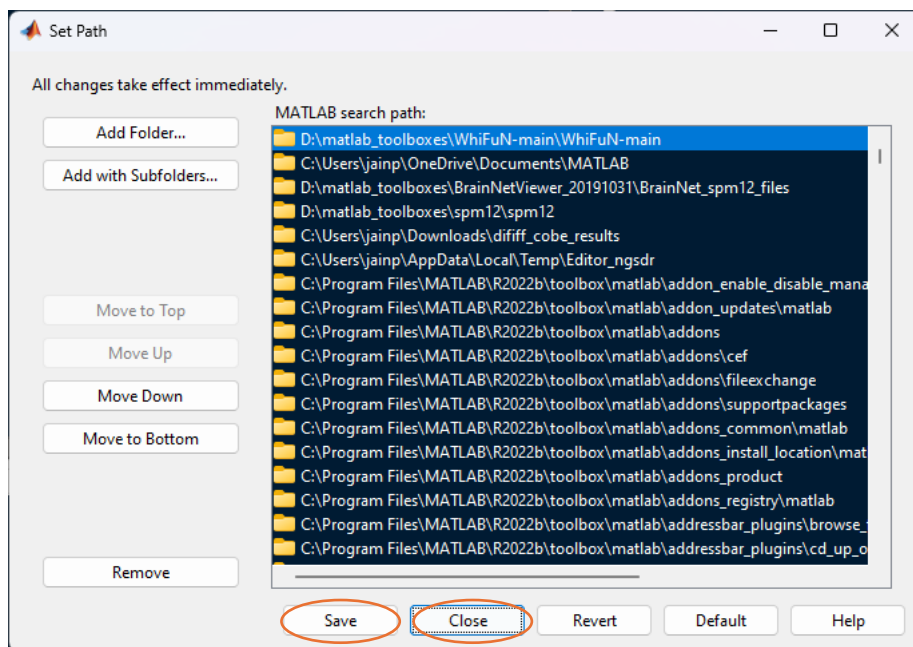
Set Path window opens up.



Click on Add folder and Select the WhiFuN folder that has the whifun.m code



You will see the folder added to the path. You can save and close the set path window.



Once all the paths are added, type whifun in the MATLAB command window and the WhiFuN GUI will pop up.

WhiFuN v1


WhiFuN

Setup

☒ BIDS

☐ All folders are Subjects

Check Data

Data Check Report

Preprocess

Pipeline	Tools
No. of Volumes to Discard 10 FD thresholds for motion outlier detection (mm) max 5 mean 0.2 Greater than 20% 0.2 Nuisance Regression Mean CSF ☒ 24 Motion Parameters	Temporal Filtering (Hz) ☒ 0.01 ~ 0.15 Smoothing WM-GM Separate FWHM (mm) 4 Normalization (voxel dim (mm)) 3 ☐ Overwrite existing files Run Preprocessing

Construct FN and FC

Check the Quality Control Plots and exclude subjects with bad data
☐ All Subjects are good or ☐ Manually exclude subjects Selected Subjects with good data

WM-FN	GM-FN	Atlas-FC	Extract Time Series in frequency band
Individual WM Probability threshold 0.6	Group WM threshold (%) 100		
K-means (WM) (K-Range) 2 ~ 22	Num CV folds 4		
Sub-sampling the WM voxel FC matrix Let toolbox decide			
☐ Corpus Callosum Networks		Create WM-FN	

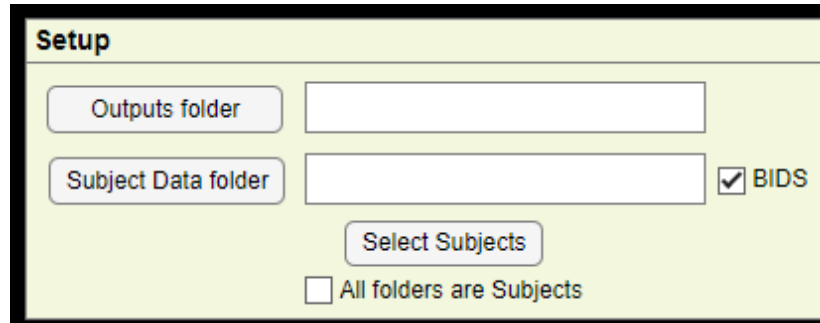
Analyze FN and FC

Visualize

Save/Load Parameters

2. Setup

The Folder paths for where the MRI data is stored and where the outputs of WhiFuN will be stored is provided in the Setup.



The Setup window contains the following elements:

- Outputs folder:** A button next to a text input field.
- Subject Data folder:** A button next to a text input field.
- BIDS:** A checked checkbox next to the text 'BIDS'.
- Select Subjects:** A button.
- All folders are Subjects:** An unchecked checkbox next to the text 'All folders are Subjects'.

2.1 Output Folder

Output Folder is an empty folder where the quality control plots and results of WhiFuN will be stored.

Click the Outputs folder button or alternatively paste the path of the Outputs folder in the text box besides the Output folder button. This will be referred to as `Output_folder_path` for the rest of the manual.

2.2 Subject Data Folder

Subject Data Folder is where the raw MRI data is present. This folder should contain subfolders that correspond to data of Subjects. Every subfolder should contain anatomical MRI scan and the fMRI scan.

Similar to the Output folder, the Subject Data Folder path can be specified by clicking the Subject Data folder button or by pasting the path in the text box besides the Subject Data Folder button. Once the Subject Data Folder is selected, WhiFuN will tell the number of folders identified in the Subject Data Folder.

If all the folders inside the Subject Data Folder correspond to a subject, then check the *All folders are Subjects* checkbox. Else, the *Select Subjects* button can be used to select the folders corresponding to the Subjects. Once the subjects are selected, WhiFuN will show the total number of subjects selected.

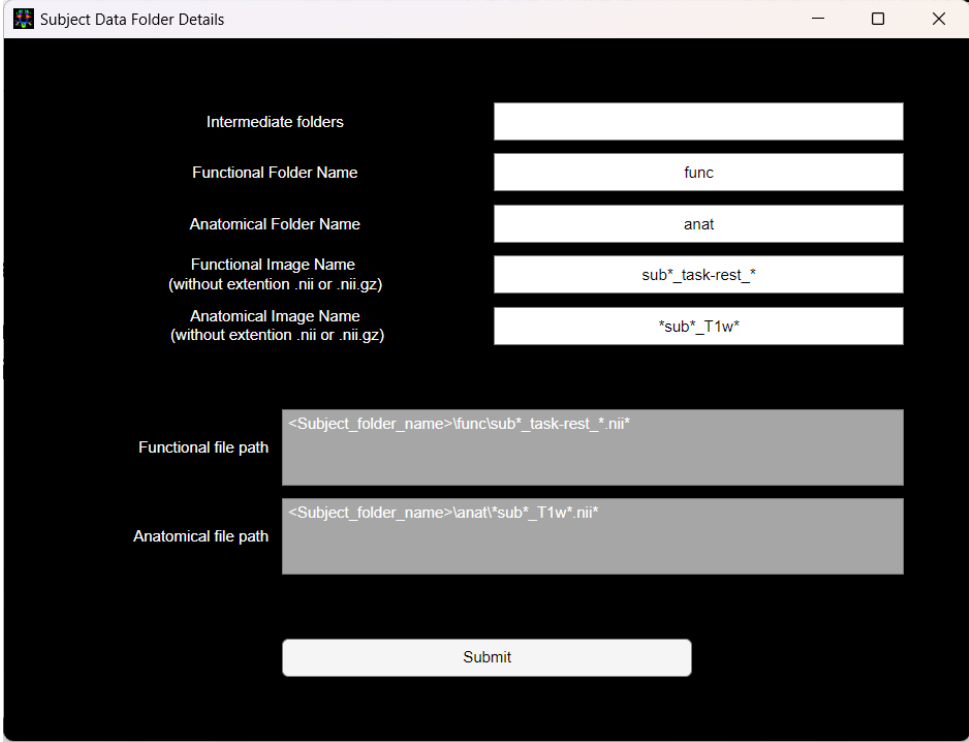
As an example, the practice data uploaded here

(<https://drive.google.com/drive/folders/1l7dhG8dYYRCW5EWhkPZbBpA7TOau1W-B?usp=sharing>) can be used to understand the paths to be given to WhiFuN.

2.3 Data Structure

Most fMRI datasets use the Brain Imaging Data Structure (BIDS) (<https://bids.neuroimaging.io/>) . If the dataset that is to be preprocessed and analyzed is in BIDS format, then the checkbox besides the Subject Data Folder should be checked.

If, however, the dataset is not in BIDS format then uncheck the BIDS checkbox. That will open *Subject Data Folder Details* window. By default, the fields here are filled for the BIDS format. To use any other data format these fields have to be changed.



Subject Data Folder Details

Intermediate folders

Functional Folder Name

Anatomical Folder Name

Functional Image Name
(without extension .nii or .nii.gz)

Anatomical Image Name
(without extension .nii or .nii.gz)

Functional file path

Anatomical file path

Submit

WhiFuN can process the following deviation from BIDS format

1. The dataset to be processed should have subject folders and every subject folder should have an anatomical folder which contains the anatomical MRI scan (e.g. T1 weighted scan) and a functional folder which contains the functional MRI scan (e.g. Resting_state scan).
2. The anatomical and the functional folder should be in the same folder.

WhiFuN won't be able to process any dataset that deviates from the above-mentioned format.

We explain how the deviations from BIDS can be handled by WhiFuN using an example of practice [ABIDE](#) dataset that can be downloaded from [here](#). It has the following folder structure

Example Data format 1:

practice_nyu_abide	Subject Data folder
0050952	Subject Folder
session_1	Intermediate folder
anat_1	Anatomical folder
rest_1	Functional folder
0050953	
session_1	
anat_1	
rest_1	
0050954	
session_1	
anat_1	
rest_1	
0050955	
session_1	
anat_1	
rest_1	

Fields of *Subject Data Folder Details*

Intermediate folders: If there are folders between the subject folder and the folder containing the anatomical and functional images then those should be specified here.

In the practice dataset example (example format 1) , the intermediate folder field will have

Intermediate folders: - session_1

Note that if there are slight differences in the intermediate folders across subjects then the wild card can be used. (For e.g. Session*).

If there are multiple folders, then the complete path between the subject folder and the folder that contains the anatomical and the functional folder should be provided.

For example, if the data has the following structure

Example Data format 2

Practice_nyu_abide

```
0050952
  MRI
    3T
      session_1
        anat_1
        func_1
0050953
  MRI
    3T
      session_1
        anat_1
        func_1
```

For example, data format 2 the Intermediate folders field should be

Intermediate folders: - MRI/3T/session_1 (for windows)

or

Intermediate folders: - MRI\3T\session_1 (for Mac or Linux)

Users can also use wild cards (*).

Functional Folder Name: Specify the folder name where the functional file is present.

For example: for both the example data formats, the functional folder name will be

Functional folder name: - func_1

Anatomical folder name: Specify the folder name where the anatomical file is present.

For example: for both the example data formats, the anatomical folder name will be

Anatomical folder name: - anat_1

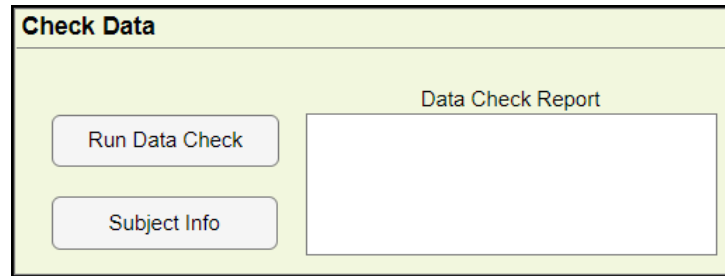
Functional/Anatomical Image name: Specify the name of the functional/anatomical image here. Since most datasets have the subject ID included in the functional image name, one would have to use the wild card (*) here.

Most datasets have the compressed files in the form *.nii.gz*, while others have the *.nii* files. Do not specify the extension in the field, WhiFuN will automatically detect the file and if a *.nii.gz* file is found then it will uncompress it and process the file.

Once all the fields are set, the path of the functional and anatomical images can be seen in the functional and anatomical file path fields. The user must check these paths and then click submit. As the user clicks submit, WhiFuN will check if the given anatomical and functional path works for the first subject, if it does not, then it will say which field it feels can be wrong. The user can do the following changes and try again.

3. Check Data

Once the Setup is complete WhiFuN will check all the paths provided and inform the user of any subject that has missing anatomical or functional files. Additionally, the anatomical and functional voxel sizes, number of time points and Time resolution (TR) of the functional image will also be checked.

The image shows a software interface titled "Check Data". It has a light green background. On the left side, there are two buttons: "Run Data Check" and "Subject Info". On the right side, there is a large white rectangular area labeled "Data Check Report".

Check Data	
Run Data Check Subject Info	Data Check Report

3.1 Run Data Check

Once the Setup is complete the user must click this button to start the data check. This might take some time depending on the number of subjects. Once the check is done, Two plots will be saved in the folder <Output_folder_path>/Quality_Control

Scanning Parameters

Q1a_scanning_parameters: A bar plot of the number of fMRI/functional images/volumes, TR of the functional image, and voxel sizes of the functional and structural image is saved. The user must check that the number of fMRI images/volumes, the TR of the fMRI images and the voxel sizes must be same across the subjects

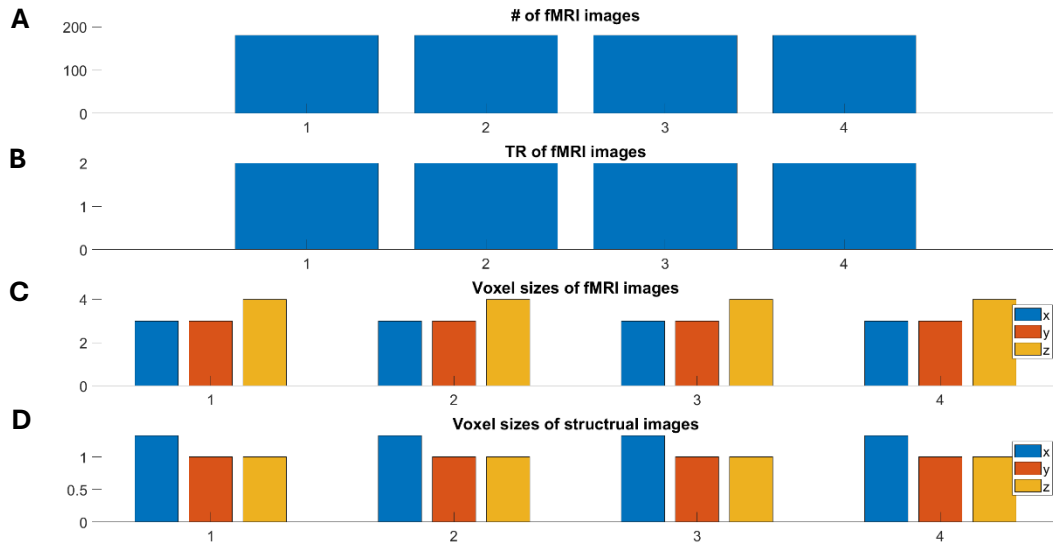


Figure 3-1: Q1a_scanning_parameters.jpg saved when the practice data set was used. The x axis denotes the subjects. A: Number of fMRI images/volumes B: TR C: Voxel size of fMRI images D: Voxel sizes of structural images, across all subjects.

Scanning parameters Histogram

If the number of subjects is a lot, then the Scanning parameters histogram would be a better choice to see if the MR parameters are consistent across the subjects. Here the user must check that there should be just one histogram bar for every plot. If there are more than one bar for a histogram (e.g. See Figure 3-1 D, voxel x dimension of structural MRI image) that means, that particular parameter is not consistent across subjects. In this case the difference is very small (1.329995 and 1.330005) users may ignore this as the difference is very small. But if the difference is large than that particular subject may be discarded.

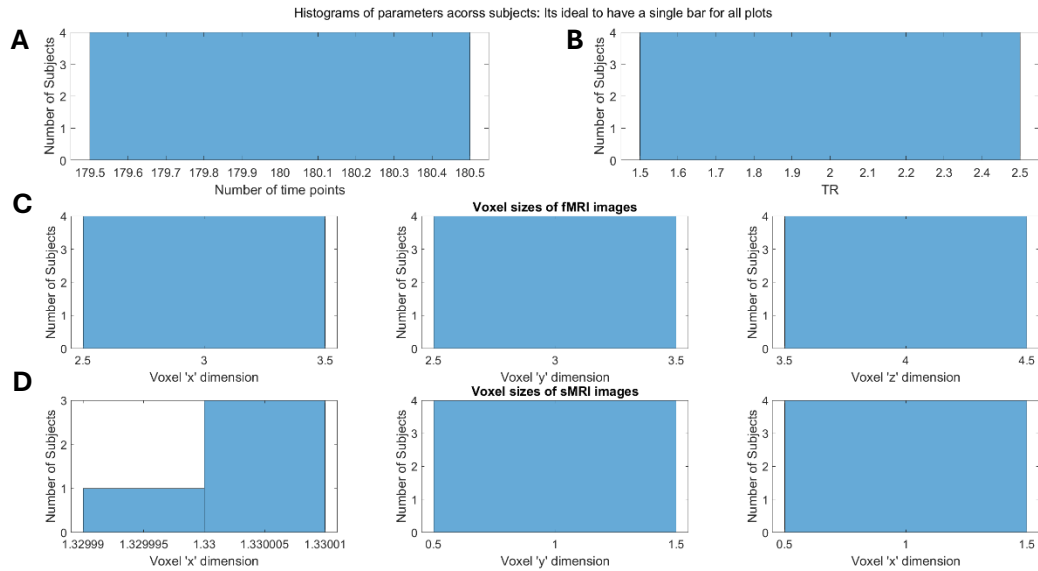


Figure 3-2: Q1a_scanning_parameters_histogram.jpg saved when the practice data set was used. Histogram of A: the number of timepoints/volumes/images in the fMRI image. B: TR C: Voxel sizes of fMRI images D: Voxel sizes of structural images.

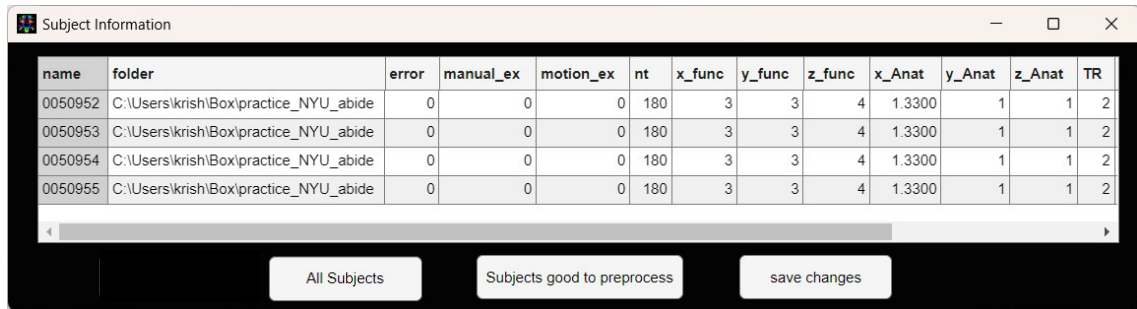
Data check Report

The data check report mentions the information of any missing files. If the check completes without any problems, Data Check report will state that the Data Check was completed successfully.

WhiFuN reads the TR value from the nifti header. There may be instances where the nifti header is corrupted or the TR value is not present in the nifti header. Data check report will notify the user that the TR corresponding to Subject <Subject ID> is not found and the user will have to manually specify the TR for those subjects.

3.2 Subject Info

Once the Data check is done, User can look at the information extracted from WhiFuN using the Subject Info button.



name	folder	error	manual_ex	motion_ex	nt	x_func	y_func	z_func	x_Anat	y_Anat	z_Anat	TR
0050952	C:\Users\krish\Box\practice_NYU_abide	0	0	0	180	3	3	4	1.3300	1	1	2
0050953	C:\Users\krish\Box\practice_NYU_abide	0	0	0	180	3	3	4	1.3300	1	1	2
0050954	C:\Users\krish\Box\practice_NYU_abide	0	0	0	180	3	3	4	1.3300	1	1	2
0050955	C:\Users\krish\Box\practice_NYU_abide	0	0	0	180	3	3	4	1.3300	1	1	2

Figure 3-3: Subject Information extracted from Data check.

The fields in the Subject Info are explained as follows

- 1) Name: The Subject name/ID same as the Subject Folder name.
- 2) Folder: The Subject Data Folder
- 3) Error: If any error was found for a particular subject, this field will be 1 corresponding to that subject else 0.
- 4) manual_ex: If any subject was excluded from preprocessing during the setup stage than manual_ex will be 1 for that subject else 0.
- 5) motion_ex: During Preprocessing if a particular subject does not meet the motion criteria then motion_ex will be 1 for that subject else 0.
- 6) Nt: Number of time points/ volumes in the functional image.
- 7) x_func, y_func, z_func: The x,y,z voxel size of the functional image.
- 8) x_anat, Y_anat, z_anat: The x,y,z voxel size of the anatomical image.
- 9) TR: The temporal resolution (TR) of the functional image extracted from the Nifti header.

Subjects good to Preprocess

If the Subjects good to preprocess button is clicked, only the subjects that are actually going to be processed will be displayed and not all.

Manual corrections and Save Changes

If the Data Report says the TR was not identified for a subject, then then user can make changes by clicking on the TR column of that subject and manual enter the TR value. Once the TR value is entered click on save changes and the TR value will be stored.

4. Preprocess

WhiFuN uses a combination of inhouse MATLAB scripts and SPM12 toolbox to preprocess the functional images. Briefly the following steps are executed for preprocessing the images (See Figure 4-1). The functional images are realigned to the first volume, motion parameters are computed, and subjects with excessive motion are discarded using the framewise displacement criteria. Then, the realigned functional images are co-registered with the anatomical image, and the anatomical images are segmented. The segmentation generates a CSF tissue probability image from which a CSF mask is generated and used to identify the CSF voxels in the functional images. The mean CSF signal is regressed from the entire brain along with Friston's 24 motion parameters. After regression, a bandpass filter with a default range of 0.01 – 0.15 Hz filters every voxel time series. The brain's GM and WM regions are then smoothed separately and are finally normalized to standard MNI space. Quality control measures are saved as image files for every subject, which can be used to check if the processing was correct.

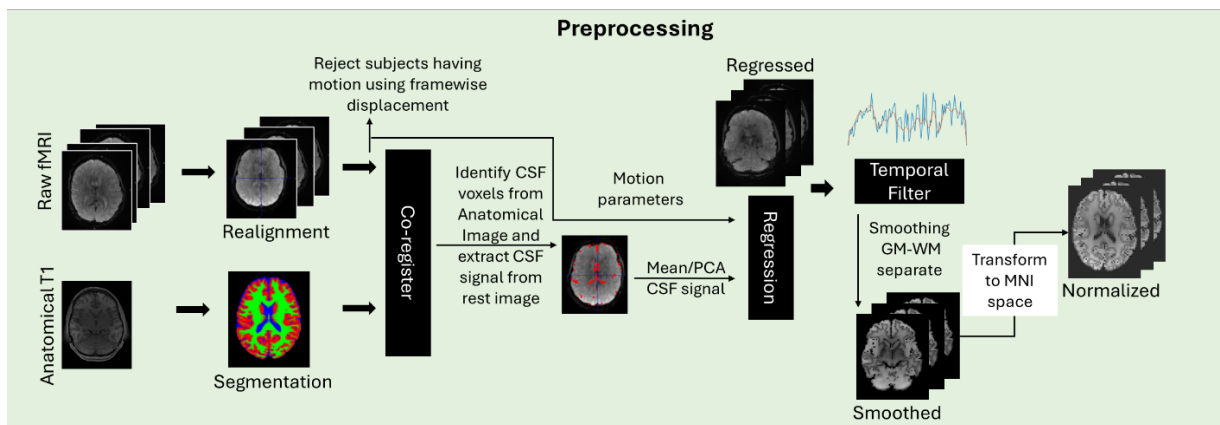


Figure 4-1: WhiFuN Preprocessing Pipeline.

Figure 4-2 shows the Preprocessing GUI of WhiFuN. The parameters related to the different preprocessing steps can be tuned here.

The screenshot shows the 'Preprocess' GUI with two tabs: 'Pipeline' and 'Tools'. The 'Pipeline' tab is active, displaying various parameters for preprocessing. The parameters are organized into sections: 'No. of Volumes to Discard' (10), 'FD thresholds for motion outlier detection (mm)' (max: 5, mean: 0.2, Greater than 20%: 0.2), 'Nuisance Regression' (Mean CSF, 24 Motion Parameters), 'Temporal Filtering (Hz)' (checked, 0.01 to 0.15), 'Smoothing' (WM-GM Separate, FWHM (mm): 4), 'Normalization (voxel dim (mm))' (3), and 'Overwrite existing files' (unchecked). A 'Run Preprocessing' button is located at the bottom right.

Figure 4-2: WhiFuN Preprocessing GUI

In the following sections we will go through the preprocessing pipeline in detail and discuss these parameters in detail.

4.1 Overwrite Existing files

This checkbox can be used to overwrite any previous files created during preprocessing.

If overwrite existing files checkbox is unchecked, WhiFuN will check if any preprocessing files already exist and skip preprocessing for these files.

If overwrite exiting files checkbox is checked, WhiFuN will delete any previously created preprocessing files (if they exist) and start the preprocessing from the start for every subject.

4.2 Unzipping and Discarding initial Volumes

The preprocessing pipeline starts by '*gun zipping*' any .gz file and discarding initial volumes (default is 10) from the functional images, allowing for the magnetization to stabilize to a steady state (Caballero-Gaudes & Reynolds, 2017).

No. of Volumes to Discard → Expects positive integer values including 0.

If the user specifies 0, No initial volumes will be discarded.

Output of Discarding initial Volumes

- 1) c_<functional image name>.nii :- Image with the initial volumes discarded as specified by the user. This file will be stored in the folder that contains the functional image.

If 0 initial volumes were specified to be discarded by the user then, no new image will be generated.

4.3 Realignment and Framewise Displacement

The raw functional volumes are realigned to the first image using SPM's '*Realign (estimate and reslice)*'. The head motion associated with each volume is estimated by evaluating how much each volume is transformed to match the first volume.

The following files will be stored in the folder that has the functional file after Realignment is successfully completed.

Outputs of Realignment

- 1) r<cut functional image name>.nii :- Realigned image generated after discarding the initial volumes.
- 2) Rp_<cut functional image name>.mat :- Motion parameters namely the translations of each volume in x, y, and z directions in millimeters and the rotations of each volume in pitch, roll, and yaw in radians in a text file.
- 3) mean<cut functional image name>.nii :- Mean functional image is a 3D image with every voxel representing the mean value of the BOLD timeseries.

Framewise Displacement

Using the motion parameters obtained from Realignment the Framewise Displacement (FD) is calculated using the equation proposed by Power and colleagues (Power et al., 2012).

$$FD = (x_t - x_{t-1}) + (y_t - y_{t-1}) + (z_t - z_{t-1}) + 50((\alpha_t - \alpha_{t-1}) + (\beta_t - \beta_{t-1}) + (\gamma_t - \gamma_{t-1}))$$

Where x_t, y_t, z_t is the translation motion (in cm) in the x, y and z direction and $\alpha_t, \beta_t, \gamma_t$ are the rotational motion parameters (in radians) at timepoint t . Here, it is assumed that the head is a sphere with a radius of 50 cm and that the axis of rotation passes through the center of the head. This gives $nT - 1$ FD value for where nT is the total number of time points/volumes in the functional image.

Outlier volumes are identified and corresponding scans are excluded if the maximum FD is greater than a threshold (default is 5mm), if the overall mean FD is greater than a threshold (default 0.2 mm), or if more than 20% of volumes are greater than a threshold (default 0.2 mm) (Parkes et al., 2018).

Once the preprocessing is completed for all subjects, based on the distribution of the FD values WhiFuN allows the user to set the above-mentioned thresholds based on how much motion can be tolerated for a particular cohort (See Section 6.2).

4.4 Segmentation, Co-registration and Mask Extractions

After realignment, WhiFuN performs segmentation and skull stripping on the anatomical image. Next, the functional images are co-registered to the skull-stripped anatomical image to ensure spatial alignment. A cerebrospinal fluid (CSF) mask is then generated, and the BOLD time series within the CSF mask is extracted from the functional images. There are no parameters that the user must define for these steps.

Output of Segmentation

Segmentation generates the following files in the folder that has anatomical image.

Tissue Probability Maps in Subject Space

- 1) c1<anat image name>.nii :- The GM tissue probability map in subject space. It is a 3-dimensional image with the same size as that of the anatomical image, with voxel values representing the probability of it belonging to GM.
- 2) c2<anat image name>.nii :- The WM tissue probability map in subject space. It is a 3-dimensional image with the same size as that of the anatomical image, with voxel values representing the probability of it belonging to WM.
- 3) c3<anat image name>.nii :- The CSF tissue probability map in subject space. It is a 3-dimensional image with the same size as that of the anatomical image, with voxel values representing the probability of it belonging to CSF.

Tissue Probability Maps in Subject Space

- 4) wc1<anat image name>.nii :- The GM tissue probability map in MNI space. It is a 3-dimensional image with the same size as that of the MNI template, with voxel values representing the probability of it belonging to GM.

- 5) wc2<anat image name>.nii :- The WM tissue probability map in MNI space. It is a 3-dimensional image with the same size as that of the MNI template, with voxel values representing the probability of it belonging to WM.
- 6) wc3<anat image name>.nii :- The CSF tissue probability map in MNI space. It is a 3-dimensional image with the same size as that of the MNI template, with voxel values representing the probability of it belonging to CSF.

WhiFuN also saves the modulated versions of the above mentioned images (mwc*<anat image name>.nii , where * = 1,2 or 3 for GM, WM or CSF.

Files required for Normalization to MNI space.

- 7) y_<anat image name>.nii :- Deformation field. This file contains the transformation parameters needed to warp images from the subject's native space to standard MNI space. It is used for normalizing any image to the MNI template.
- 8) iy_<anat image name>.nii :- Inverse Deformation field. This file contains the inverse transformation parameters, allowing images in MNI space to be mapped back to the subject's native space.

Bias Regularized Image

- 9) m<anat image name>.nii :- bias regularized image. Removes the smooth spatially varying artifact that modulates the intensity of the image (bias).

Outputs of Skull Stripping

- 1) b<anat image name>.nii :- Skull stripped Anatomical image.

Output of Co-registration

The header of the realigned functional image is changed. No new image is generated.

- 1) <Realigned functional image name>.mat :- A MATLAB variable file, having the new transformation matrix information after the co-registration is done.

Outputs of Masks Extraction

- 1) anat_mask.nii :- Binary mask in subject space (same size as that of the original anatomical image) where voxels inside the brain are labeled as 1 and those outside as 0.
- 2) wanat_mask.nii :- Binary mask in MNI space (same size as that of the MNI template) where voxels inside the brain are labeled as 1 and those outside as 0.
- 3) CSF_MASK0.95.nii :- Binary mask identifying the CSF voxels in subject space. Voxels with CSF probability greater than 0.95 are labeled as 1, while others are 0.

The following file is stored in the folder where the functional file is present.

- 4) Covariance_csf_REST.mat :- This is a .mat file that contains the mean/PCA of the CSF BOLD signal which is to be regressed during the Nuisance regression step.
- 5) rest_mask.nii :- Binary mask in the subject space for the functional image where voxels inside the brain are labeled as 1 and those outside as 0.

4.5 Nuisance regression

To reduce noise, WhiFuN regresses out unwanted signals from every voxel in the brain. Users can choose between regressing the mean CSF signal for a faster computation or using principal component analysis (PCA) to regress the dominant CSF signal components, which is more effective but computationally slower. Alternatively, users may opt to skip CSF signal regression altogether.

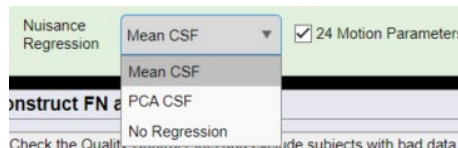


Figure 4-3: Options for Nuisance regression.

Besides the dropdown for the Nuisance regression, is the check box for regressing out the Friston's 24 motion parameters (Friston et al., 1996; Yan et al., 2013). Friston's 24 parameters are the three translational and three rotational parameters, the squares of them, the derivatives of them, and the squares of the derivatives. Regression is performed to remove the nuisance signals from the rsfMRI time series. If this box is unchecked the motion parameters won't be regressed out.

If the user decides to use the PCA of CSF signals, then WhiFuN will prompt an additional popup asking for the number of PCA components to regress.

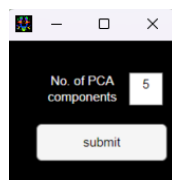


Figure 4-4: Number of PCA components to regress.

Output of the Nuisance Regression

- 1) Reg_<realigned functional image name>.nii :- Image with the specified confounds regressed.

4.6 Temporal Filtering

After Regression, the voxel time series from every voxel is filtered using a Butterworth bandpass filter of second order in the range specified by the user (default 0.01 to 0.15

Hz). The user may choose to skip the filtering step, by unchecking the check box besides Temporal filtering (Hz).



Figure 4-5: Filter frequency range

Output of the filtering

- 1) f<regressed functional image name>.nii :- filtered functional image within the specified frequency bands.

4.7 Smoothing

Smoothing is a very important step for WM analysis of BOLD signals. Some criticism in the field is that the GM BOLD signals mix into the WM signals. To completely avoid that WM and GM signals are separately smoothed. WhiFuN also allows the user to use the predefined smooth module for SPM that smooths the WM and GM together, but this is not recommended for WM analysis. Smoothing is performed using a Gaussian kernel with Full with half maximum (FWHM) that can be specified by the user. Moreover, the users may choose not to perform the smoothing step.

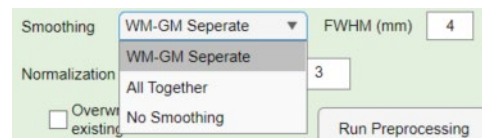


Figure 4-6: Options for Smoothing

Output of the smoothing

- 1) s<filtered functional image name>.nii :- Smoothed Image

4.8 Normalization to MNI space

Normalization is performed after smoothing, where images are transformed from the subject's native space to the standard MNI space with a voxel size specified by the user (default: 3 mm). A 3 mm voxel size is commonly used because the after preprocessing, WhiFuN constructs Functional Networks using a voxel-level FC matrix. Using a smaller voxel size (e.g., <3 mm) increases the number of voxels, which may lead to excessive memory requirements, making voxel-level FC computation challenging.

The deformation field generated during segmentation is used for normalization. The normalization step has a bounding box parameter that determines the field of view of the image with respect to the anterior commissure. The default bounding box parameter set in SPM is slightly expanded to [-90 -126 -72; 90 90 108] to ensure that all brain tissue regions are included.

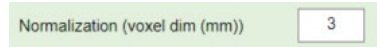


Figure 4-7: Normalization voxel dimensions.

Output of the Normalization

- 1) w<smoothed functional image name>.nii :- Image normalized in the MNI space with the voxel size specified by the user. Stored in the folder where the functional image is stored.
- 2) w<skull stripped anatomical image name>.nii :- Anatomical Image normalized in the MNI space. Stored in the folder where the anatomical image is stored.

4.9 Overwrite Existing files

If preprocessing is interrupted due to an issue such as a system shutdown or MATLAB crash, WhiFuN allows users to resume from where it stopped. Before restarting, WhiFuN checks for previously generated preprocessing files. If the required files are found, WhiFuN skips the completed steps and proceeds to the next stage. However, if the user wishes to restart preprocessing from the beginning, they must select the "Overwrite Existing Files" option. When Overwrite Existing Files checkbox is checked, WhiFuN will delete any previously generated preprocessing file and run the preprocessing from the beginning for every subject.

5. Quality Control

Once the preprocessing is complete, the user is recommended to go through the quality control plots generated after most preprocessing steps. These quality control plots are saved in <Output_folder_path>/Quality_Control.

5.1 Initial Check

The raw anatomical and functional images are stored as <Subject_ID>_anat.png, and <Subject_ID>_func.png files respectively in Initial_check folder with the image contours on the standard MNI reference template for every subject. The user must check the initial position and orientation of the images with that of the MNI space. If the anatomical image is far from the MNI template or orientated differently, then the user must reorient the image manually to the template direction and reset the origin to the anterior commissure (Di & Biswal, 2023).

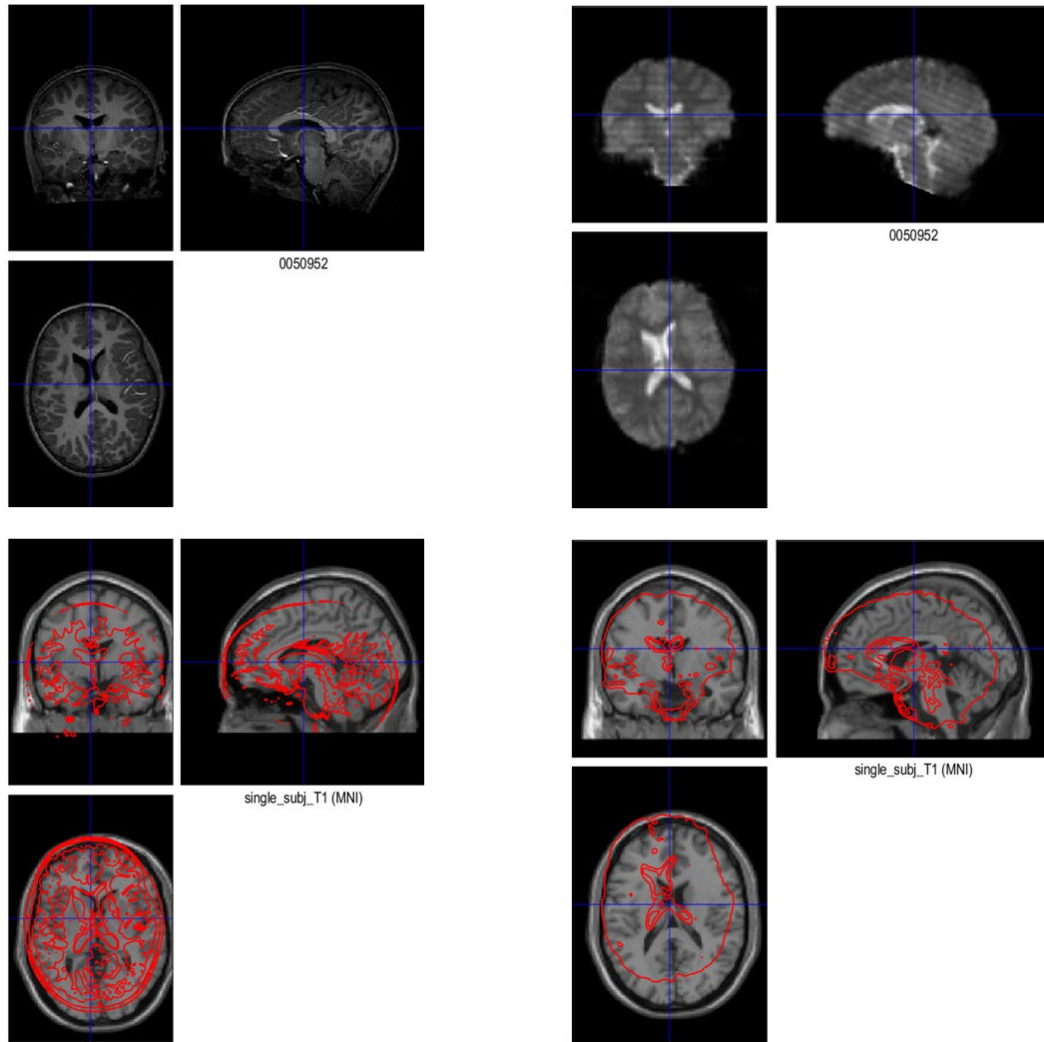


Figure 5-1: Initial Check Quality Control. The raw anatomical and functional image and their contours plotted on the standard MNI template

5.2 Head Motion

For each subject, a <Subject_ID>.png file is saved in the Head-Motion folder. This image contains a subplot with the following visualizations (See Figure 5-2):

- 1) Global BOLD Signal – Displays the overall BOLD signal across time.
- 2) Motion Parameters – Shows the translational and rotational motion parameters.
- 3) Pairwise Variance of the Global BOLD Signal – Illustrates variations in the global signal over time.
- 4) Framewise Displacement – Computed using the equation provided in Section 4.3, this plot quantifies subject movement between consecutive volumes. This plot also shows the thresholds use for head motion.

These plots allow the user to understand why WhiFuN rejected certain subjects. By default, WhiFuN automatically excludes subjects with excessive motion, requiring no additional action from the user.

It can be observed from the Figure 5-2 that the spikes in the global signal correspond to the spikes in the motion parameters. This is the reason why the motion parameters are regressed out from every voxel timeseries (See section 4.5).

Additionally, a `excluded_<Subject_ID>.txt` file is saved for each rejected subject, indicating the specific FD time points that exceeded the threshold, leading to the rejection.

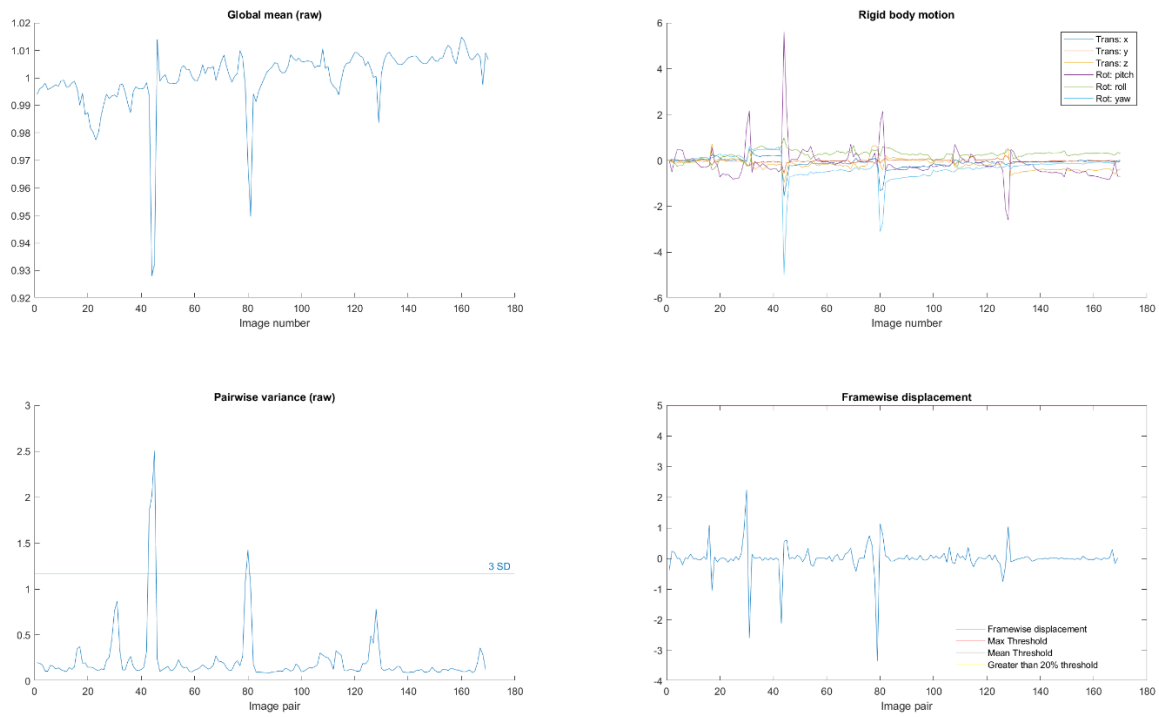


Figure 5-2: Head motion quality control plot.

5.3 Segmentation

After segmentation the GM, WM and CSF images are overlapped. WhiFuN saves the segmentation outputs with the GM, WM and CSF probabilities overlapped on the same image in red, green and blue colors respectively. It also draws the contour of the segmentation output on the reference MNI space to check if the alignment of the normalized anatomical image with respect to the MNI template is correct. Users must observe the images for each subject and note any misclassification of the tissues. If any of the tissues are misclassified, then that subject may be discarded. Also, if brain lesions or image quality issues are noticed in the anatomical image, this participant's data should be discarded (Figure 5-3).

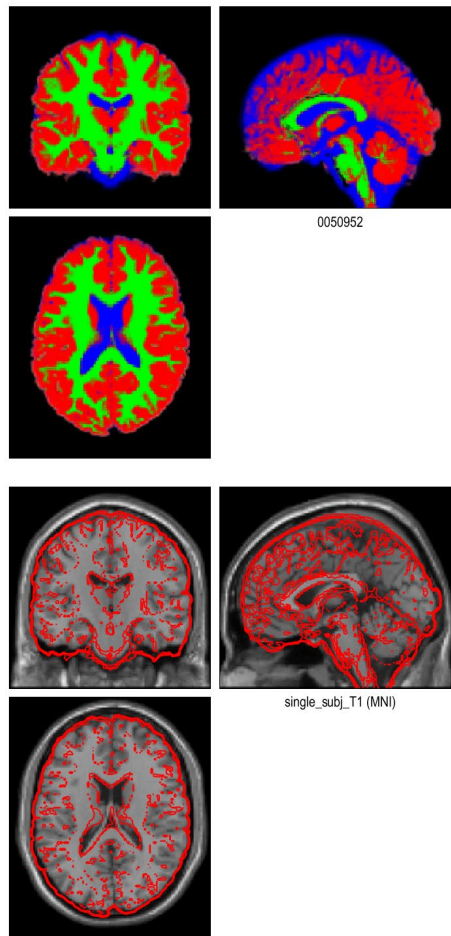


Figure 5-3: Segmentation Quality Control Plot. Gray Matter (Red), White Matter (Green) and Cerebrospinal Fluid (Blue) overlapped in the MNI space and its contour plotted on the standard MNI template.

5.4 Coregistration

The anatomical image is shown with the functional image, and the contour of the anatomical image is overlayed on the functional image in red. The user must check the orientation and alignment of the anatomical and the functional images. (See If the images are not aligned, the user can manually align them using the ‘manually coregister subjects’ module (See Section 6.2).

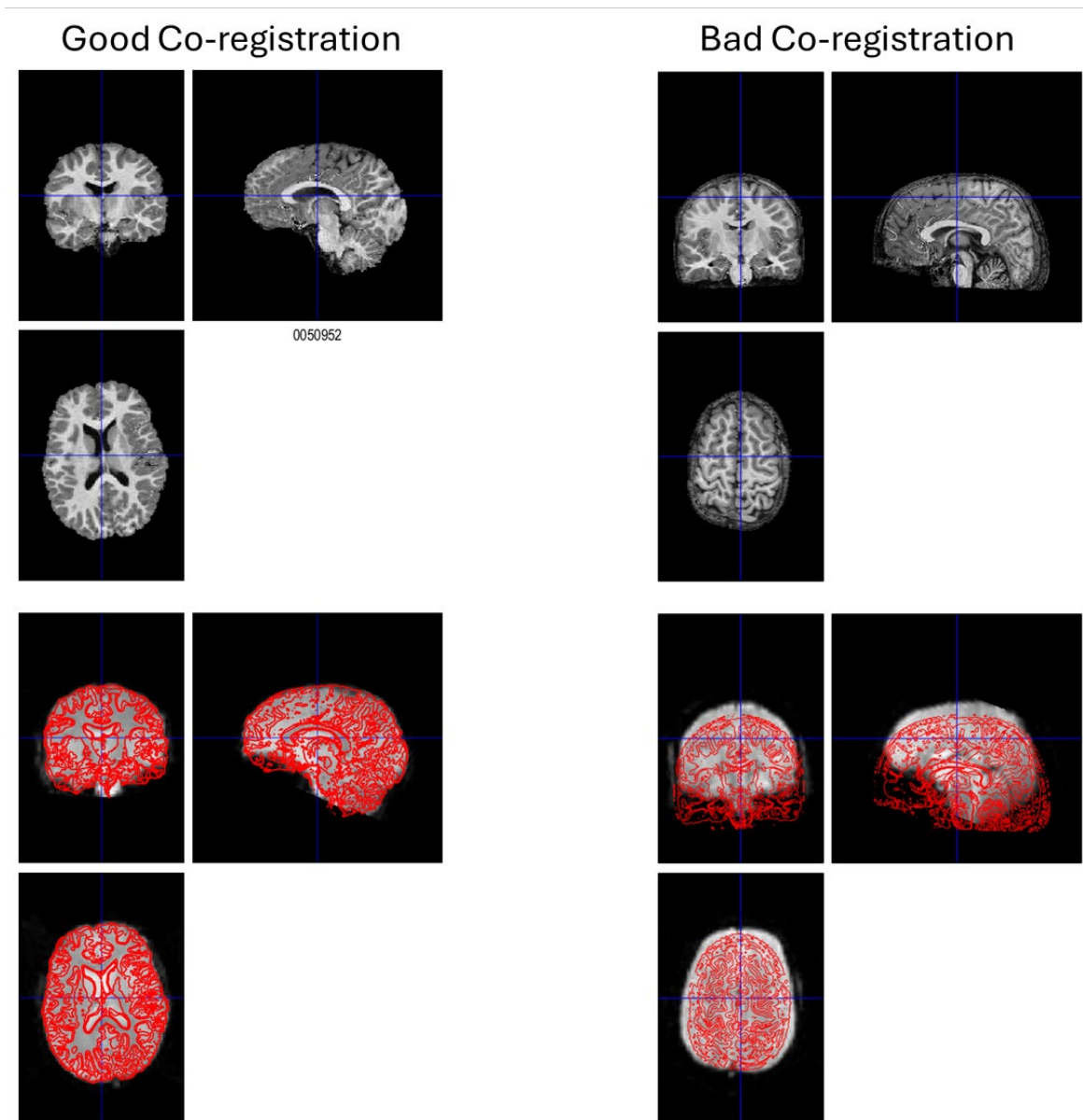


Figure 5-4 Coregistration Quality plot. Left, example of good coregistration Quality Control image. Right Example of a bad coregistered image. If the user observes an image on the right, he/she should go to section 6.2 to manually Coregister the image.

5.5 Cerebrospinal Fluid Masks

Here, the CSF mask extracted using a threshold of 0.95 on the tissue probability map of CSF is plotted with the functional image along with the contour of the CSF mask on the rsfMRI image. The user must check the alignment of the CSF mask and the rsfMRI image. It is observed in some cases that no voxels satisfy the 0.95 threshold. In such cases, the anatomical image must be rechecked, and the user may discard this subject.

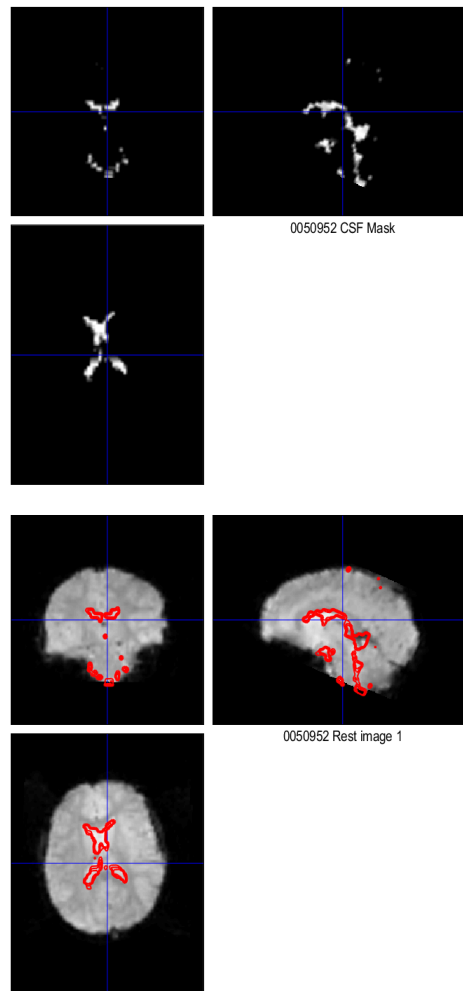


Figure 5-5: CSF mask plotted and the contour of the CSF mask overlapped on the functional image.

5.6 Effects of Regression and Temporal filtering

The global (average) signal before and after regression of the motion parameters and the CSF signals is plotted. The user may observe that any upward or downward trends and artifacts due to motion are removed after regression.

Here, the user may observe how very high or low-frequency signals are removed during filtering. These plots are just for observing the effects of regression and temporal filtering on the global signals. The user does not need to take any additional steps here.

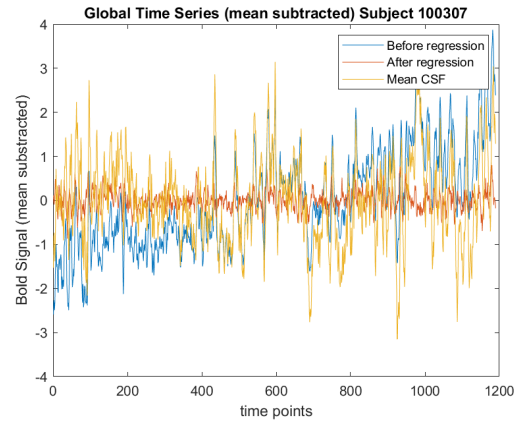


Figure 5-6 Regression Quality Control plot

5.7 Effects of Smoothing

Based on the options chosen by the user, either Smooth GM-WM separately (Recommended for WM Analysis) or Smooth all together, WhiFuN stores the smoothed image. One may observe that when the GM and WM voxels are smoothed separately, the cortex gets a number of small holes as these voxels were neither identified as GM nor WM. The accuracy of segmentation depends on the resolution of the anatomical and the rsfMRI images. If the user observes a large number of holes in the cortex, the subject may have to be discarded.

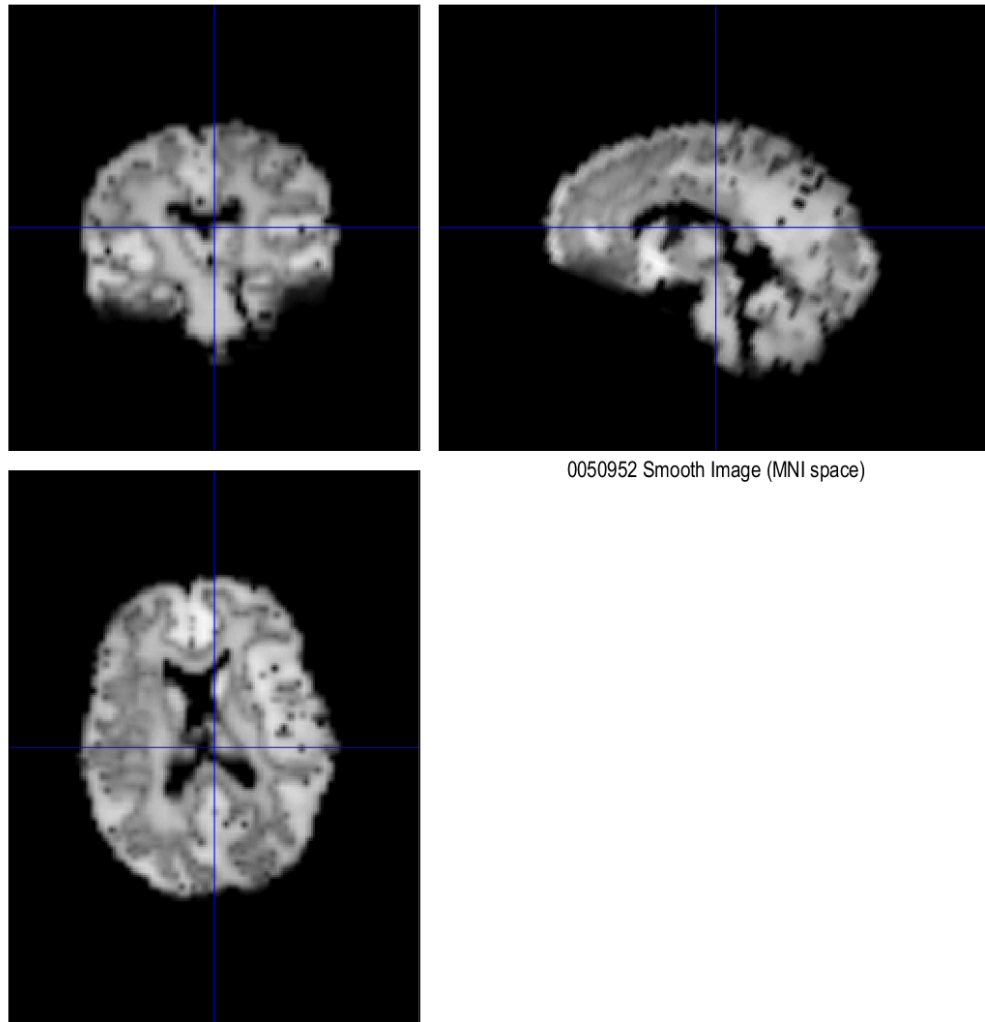


Figure 5-7: GM-WM smooth separately QC image

5.8 Normalization QC plot

The normalized functional image is plotted with the reference MNI template image with the contour of the normalized rsfMRI on the reference MNI template. The user must ensure that the spatial alignment of the normalized rsfMRI image with that of the MNI template is correct.

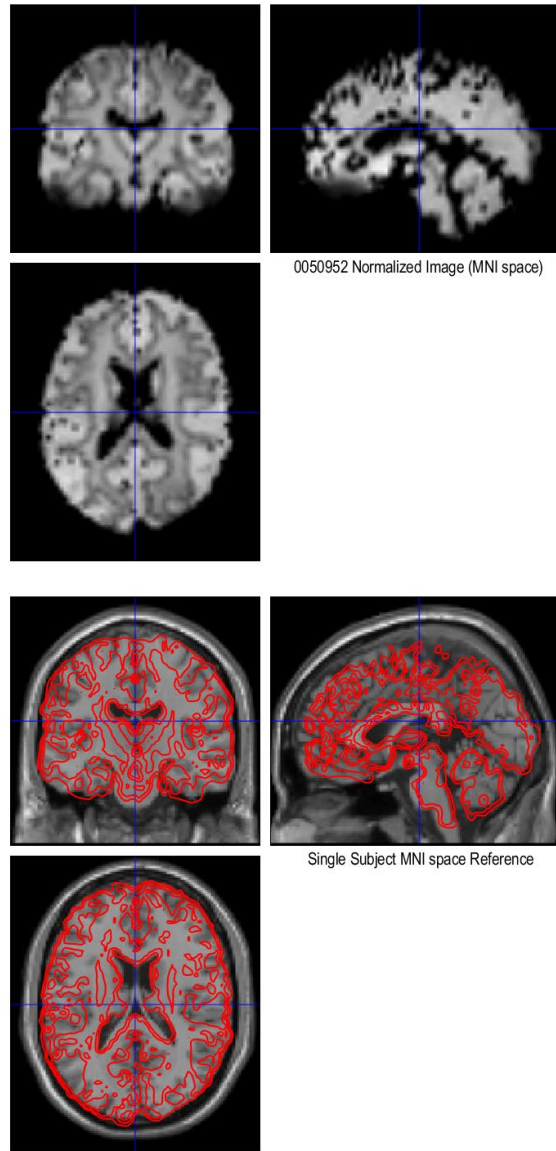


Figure 5-8: Normalization QC plot. The Normalized functional image in the MNI space is plotted with its contour overlapped on the standard MNI template.

5.9 Time-Series Check

The user can observe the associations between the global and noisy signals at this step. WhiFuN plots the raw unprocessed global signal, the six rigid motion parameters computed during realignment, the CSF signals, the global mean of the preprocessed image, and the correlation of all the above-mentioned signals. WhiFuN also plots pairwise variance between consecutive volumes (similar to DVARS (Power et al., 2014)) for the unprocessed image, the framewise displacement, the derivate of CSF signals, the pairwise variance between consecutive volumes for the preprocessed image and the correlation of these signals. The user must check the correlation plots and observe if the preprocessed global signal is correlated to the noisy signals. If the correlation of the

global signal corresponding to the final preprocessed signal with any of the noisy signals is significant, the user may exclude the subject.

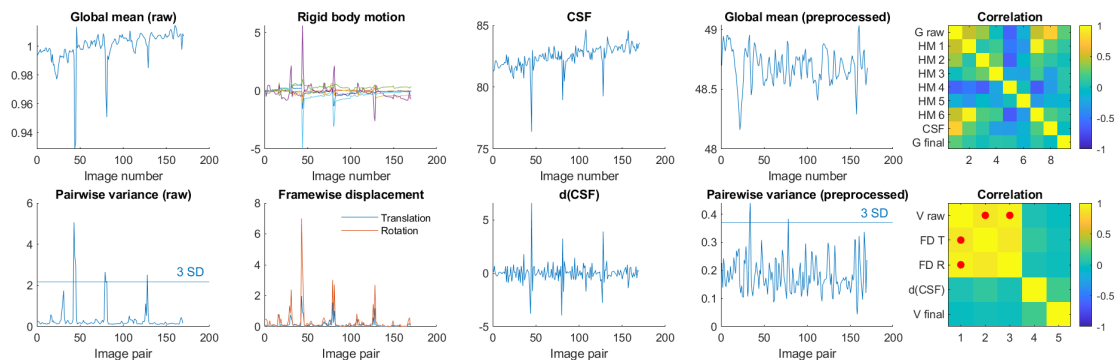


Figure 5-9 : Timeseries plot showing the global time series before preprocessing, the 6 motion parameters, the mean CSF time series, the global mean signal after preprocessing and a correlation matrix correlating these signals. The same is repeated for pairwise variance across consecutive scans instead of the global mean in the 2nd row.

5.10 Error Handling

If WhiFuN encounters any unexpected error while processing a particular subject at a particular preprocessing step, it will store the error message in a txt file named <Subject ID>_error_info.txt in the Error_info folder located in the Quality Control folder.

It is recommended that the user examines all QC plots of all the preprocessing steps to remove outlier subjects from further analysis.

6. Tools for Preprocessing

After checking the Quality Control plots the user can use the following tools to improve or redo the preprocessing if there were some errors.

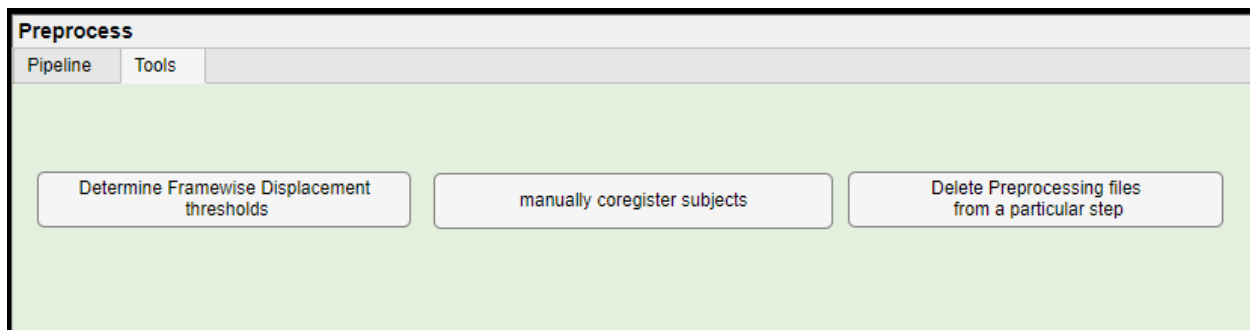


Figure 6-1: Preprocessing tools

6.1 Determine Framewise Displacement thresholds

After preprocessing, based on the motion thresholds, WhiFuN skips preprocessing steps for the subjects that had excessive motion. Thus, it may be possible that not all the subjects were preprocessed. However, the Realignment step will be done for all the subjects that were selected and the `rp_<functional_image_name>.txt` file will be generated for every subject in the functional folder. The user can see how many subjects were rejected and the reason for the rejection by using the Determine Framewise Displacement thresholds button. Once the button is clicked, a GUI shown in *Figure 6-2* opens up. This shows the Maximum FD, the Mean FD, the percentage of FD greater than 0.2 for every subject and the corresponding histograms along with the thresholds used. Using the Drop down.

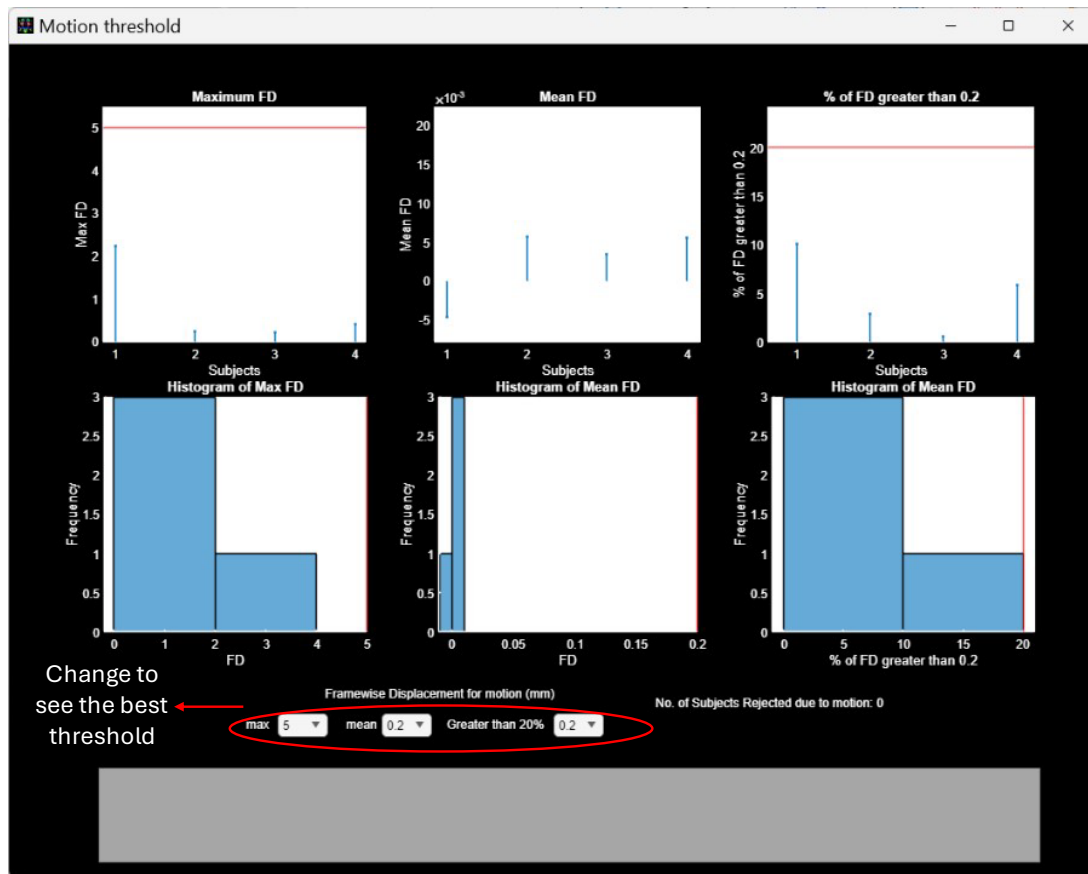


Figure 6-2: Determine the Framewise Displacement thresholds.

If a Subject is rejected as it did not pass the threshold then the subject ID and the reason why it would be rejected will be shown in the gray text box (See *Figure 6-4*).

Based on this the user can decide what threshold to keep such that optimal number of subjects can be rejected. Once the optimal thresholds are decided then the user must use these new thresholds and run the preprocessing again. The user should make sure that the Overwrite existing files checkbox is not checked (See *Figure 6-3*). WhiFuN will run the

preprocessing with the new thresholds, if the preprocessing was already completed for subjects, WhiFuN will skip the preprocessing corresponding to those subjects.

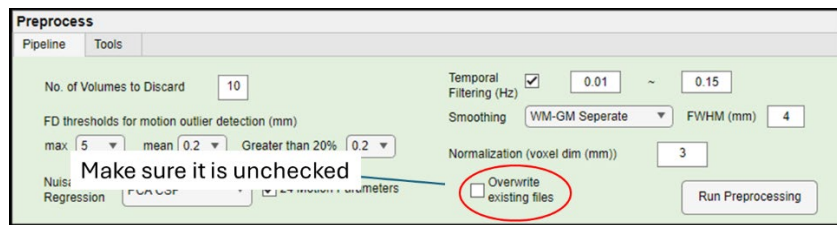


Figure 6-3 Ensure the Overwrite existing files is unchecked.

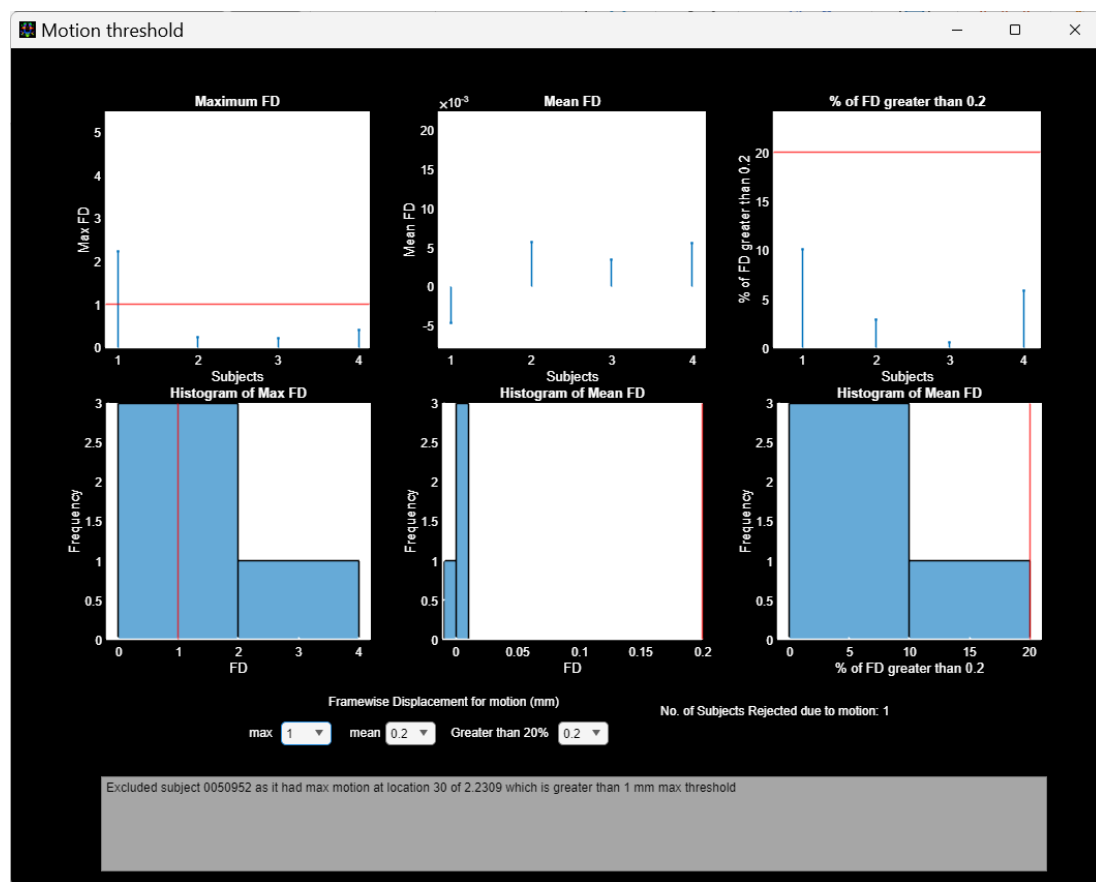


Figure 6-4 Subject Rejected as it exceeded the Max FD threshold.

6.2 Manually Coregister Subjects

If the User observes that the coregistration corresponding to few subjects is not proper (See Section 5.4) then those can be manually coregistered using this module. Once the user clicks the manually coregister subjects button, a file explorer will pop asking the user to select the subject folder corresponding to the subject that has bad coregistration (Figure 6-5)

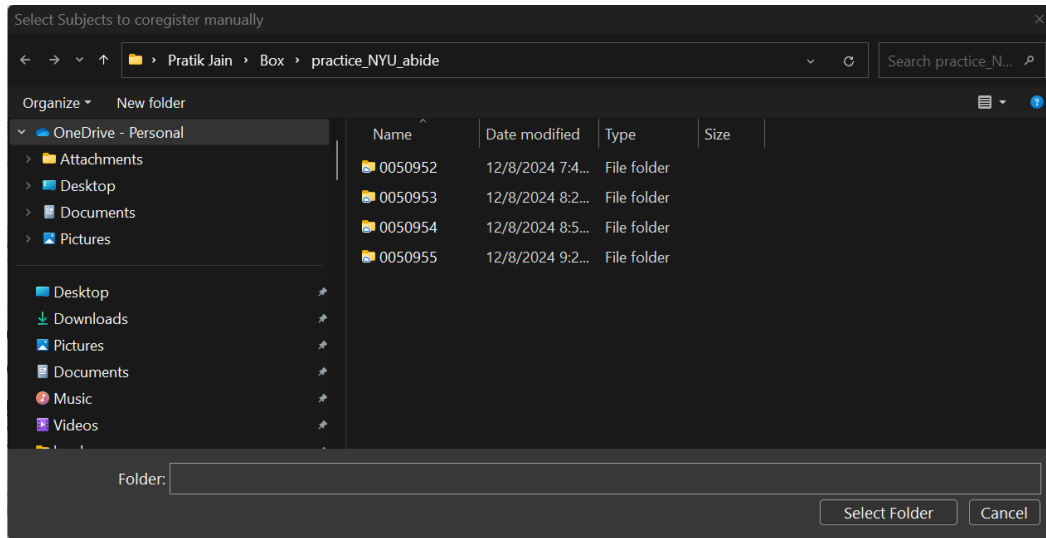


Figure 6-5: Select Subject folder that has bad coregistration.

Once the subject folder is selected, WhiFuN will prompt that the preprocessing files created after realignment steps (if exist) will be deleted. Since the Coregistration have to be performed, the preprocessing has to be redone. Once the user clicks the Yes button, WhiFuN will delete all the preprocessing that are not required and open the SPM manual coregistration module.

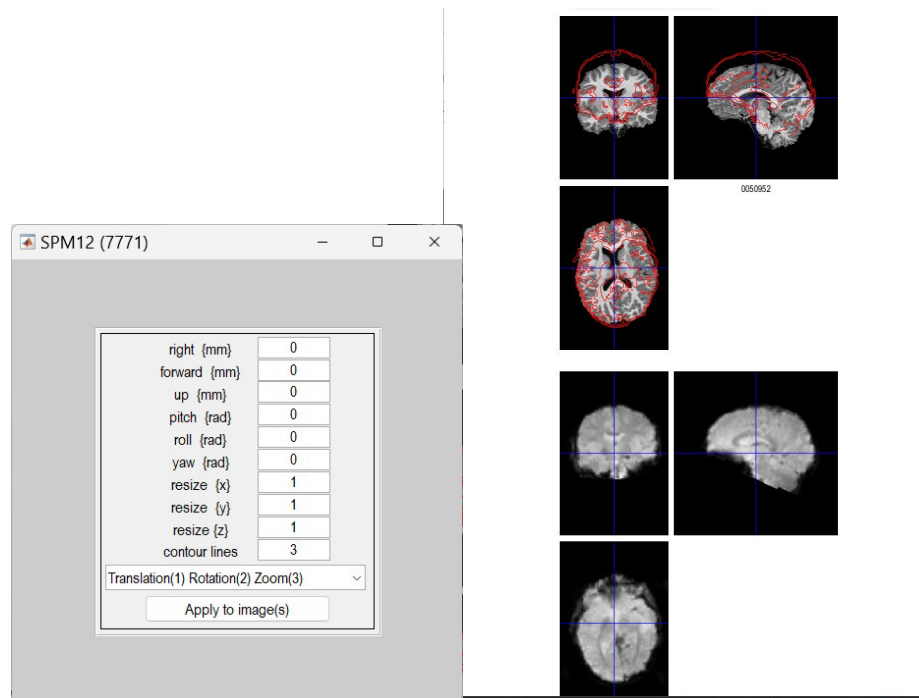


Figure 6-6: The SPM coregistration module pops up when the Subjects are selected to manually coregister.

The user can manually enter the amount of translation or rotation required so that the contour of the functional image aligns with the anatomical image.

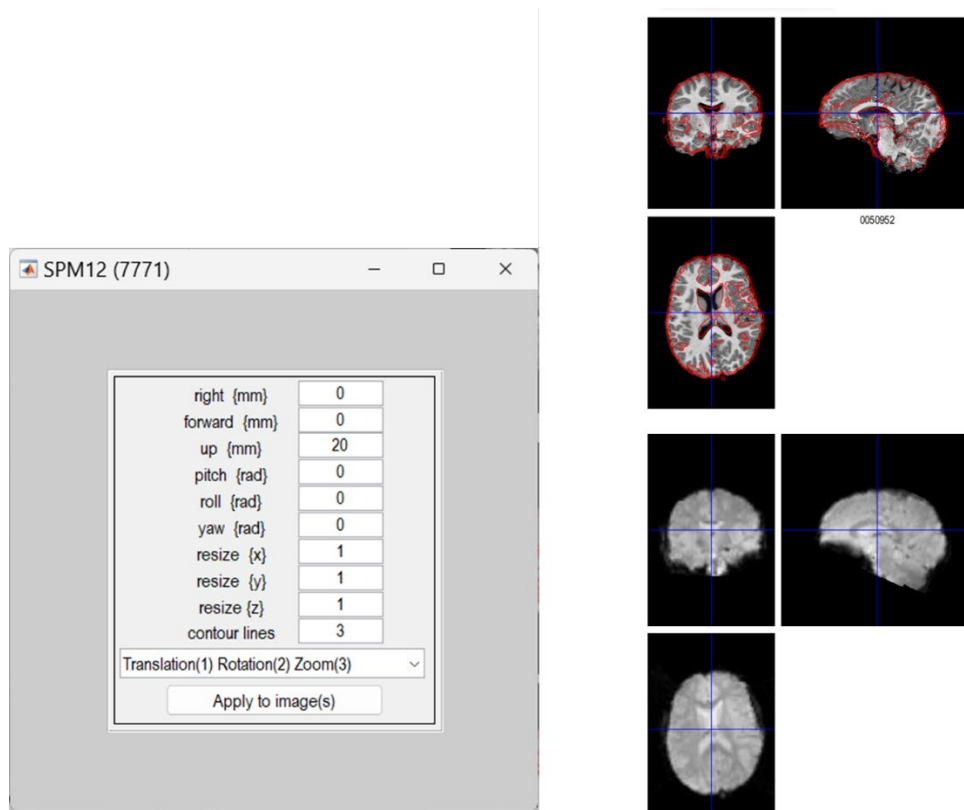


Figure 6-7: Users can manually enter the amount of translation or rotation required to align the contour of the functional image to the anatomical image.

Once the contours are aligned, the user should click the Apply to image button, which will open an SPM pop up (See *Figure 6-8*).

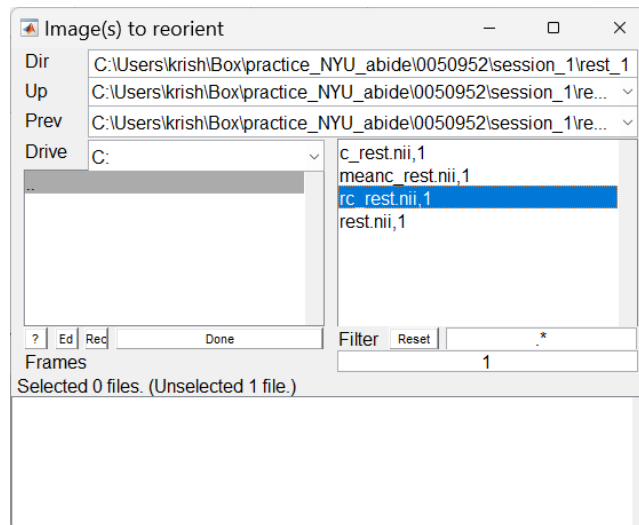


Figure 6-8: SPM pop up to apply the manual coregistration to the images

The user must select the realigned images that start with prefix 'r' if the user has also discarded some initial volumes the file name will be rc_<functional_image_name>.nii, else it will be r<functional_image_name>.nii. (in this example, <functional_image_name> = rest). Based on the preprocessing steps the prefix can be given to the filter as follows

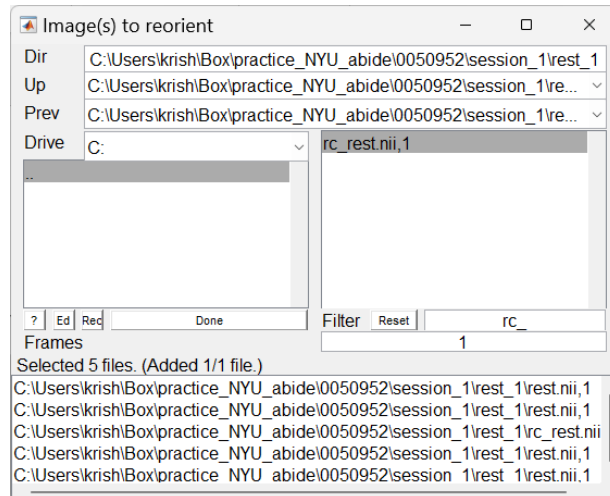


Figure 6-9: Enter the prefix of the Realigned file as filter to just select the Realigned files. Prefix = r, if no initial volumes were discarded, else if initial volumes were discarded Prefix = rc_ (default and also shown here)

It is essential to select all the functional volumes/time points, for that enter the array of volumes that has to be coregistered. In our example dataset, there are 170 time points or volumes so the array will be 1:170, In general if there are nt time points the array will be 1:nt (See Figure 6-10).

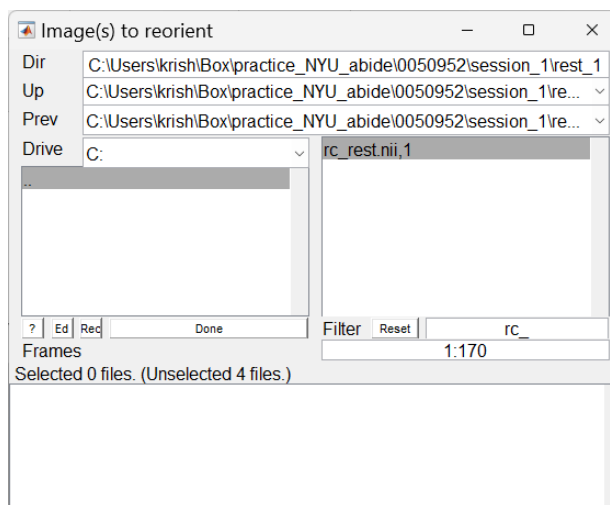


Figure 6-10: Selecting all the time points/volumes.

Then the user can use the right mouse click and select all the time points/volumes.

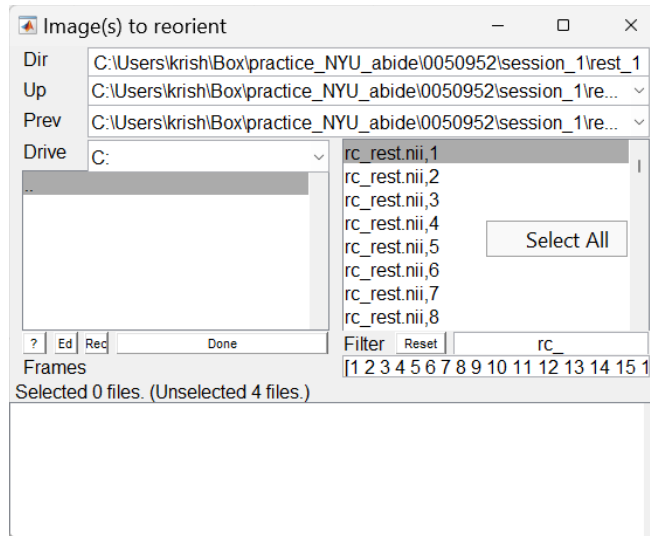


Figure 6-11: Select all time points/volumes.

Once all the files are selected, click on done and SPM will ask to save the transformation matrix (reorientation matrix), it is always advisable to save this matrix. Click yes and save the matrix in the functional folder of that subject and SPM will coregister the files.

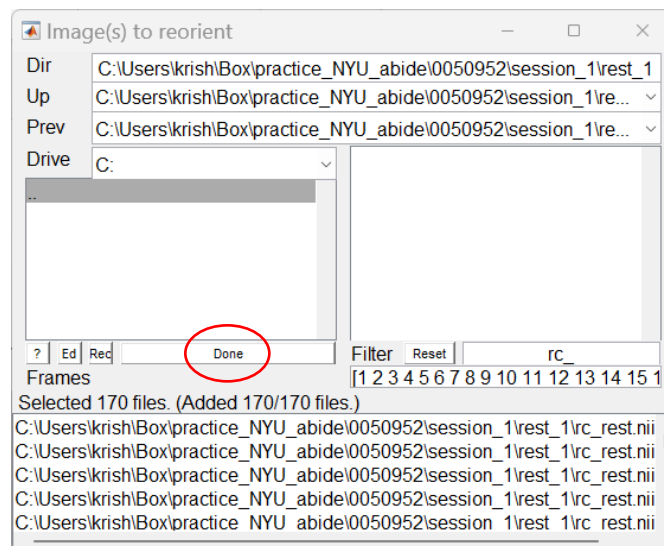


Figure 6-12: Finally Click on Done to coregister the images.

The user must ensure that the overwrite existing files check box is not checked and run the preprocessing again after this.

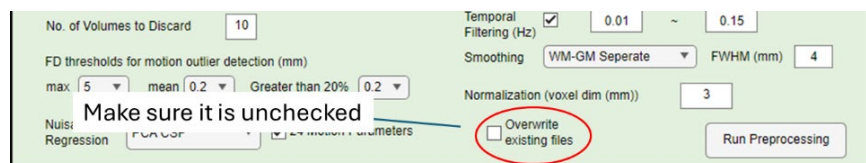


Figure 6-13 Ensure the Overwrite existing files checkbox is unchecked.

6.3 Delete Preprocessing Files from a Particular Step

If the user finds that the preprocessing was not done properly for a particular step and wishes to redo the preprocessing from that step, this module can be used. When the user clicks on Delete Preprocessing Files from a Particular Step, A GUI pops up (See Figure 6-14).

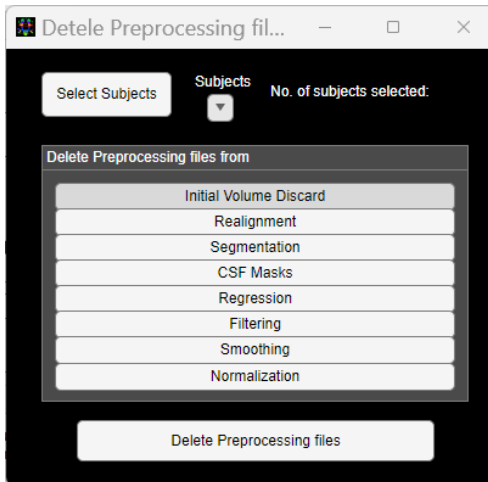


Figure 6-14: Delete Preprocessing files from a Particular step GUI.

The user will first select the subjects for which the files have to be deleted and then the step after which the files must be deleted. For instance, the following Figure 6-15 shows how all preprocessing files after CSF Masks (Including CSF Masks) are deleted for two subjects.

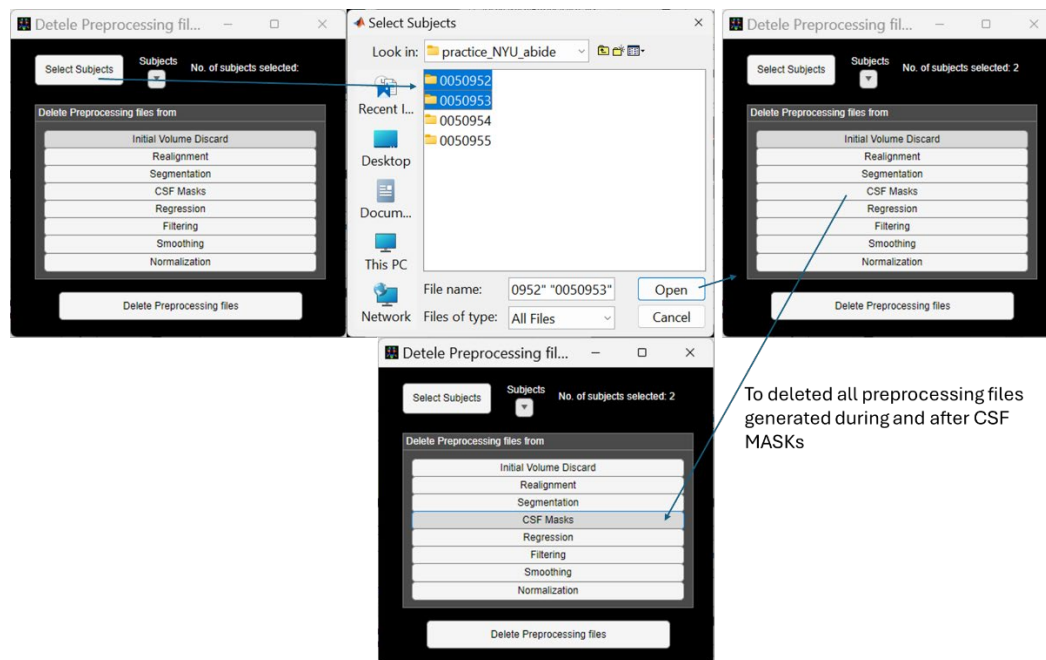


Figure 6-15: Delete Preprocessing files from a Particular step. Figure shows how preprocessing files generated during and after CSF Masks corresponding to two subjects can be deleted

7. Construct Functional Networks and Functional Connectivity

This section discusses how the Functional Networks can be extracted using the K-Means algorithm and Functional Connectivity maps can be generated by storing the average timeseries corresponding to every region in a predefined atlas supplied by the user.

7.1 Excluding the subjects with bad quality

After Preprocessing, the User must go through all the Quality controls plots described in Section 5 and reject outlier subjects. If the user detects outlier subjects, the user can discard them by clicking on Manually exclude subjects check box (See Figure 7-1).

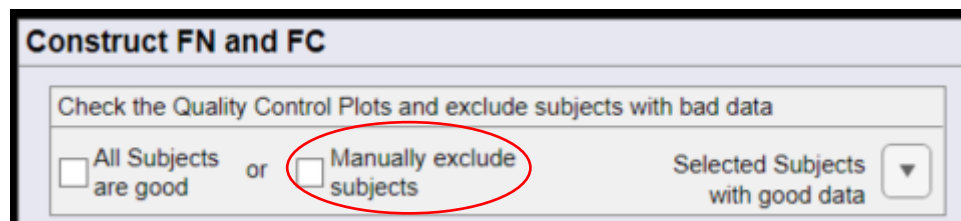


Figure 7-1: Reject subjects that have bad Quality Control plots.

Once the user checks the Manually exclude subjects check box, WhiFuN opens the subject data folder that has all the subject folders (See Figure 7-2). The User can select the subject folders corresponding to the subjects that need to be excluded due to bad quality. User can use the Ctrl button to select multiple subjects.

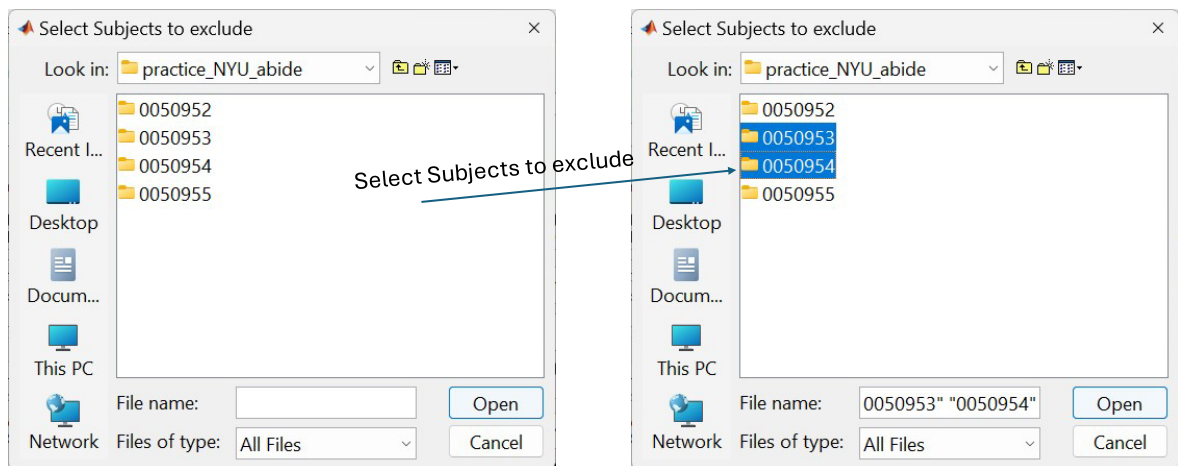


Figure 7-2: Select Subjects to exclude.

Else if all the subjects are good, click on All subjects are good checkbox (See Figure 7-3).

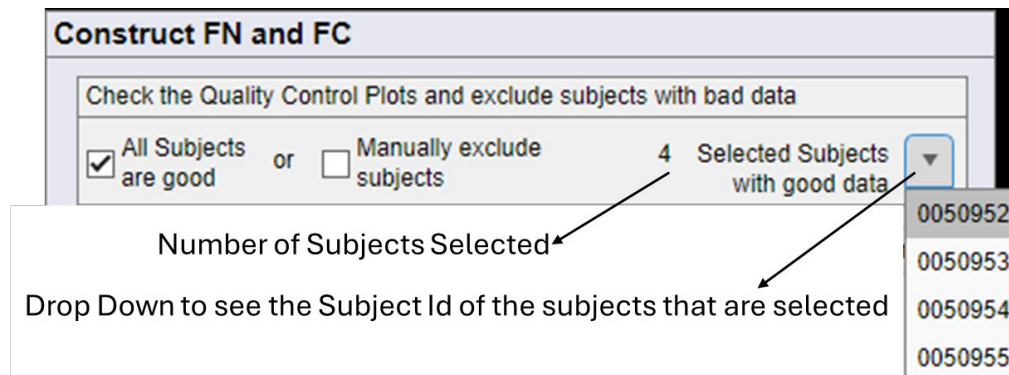


Figure 7-3: Select All subjects and other details.

7.2 White matter and Gray Matter Functional Networks

This module discusses how the WM and GM Functional Networks (FNs) can be created using WhiFuN. This module is applied to the fully preprocessed rsfMRI images. Group masks are created first to identify the WM or GM voxels across the subjects. The K-means algorithm clusters the voxels in WM and GM to create the WM and GM-FNs based on voxel-based FC (See *Figure 7-4*).

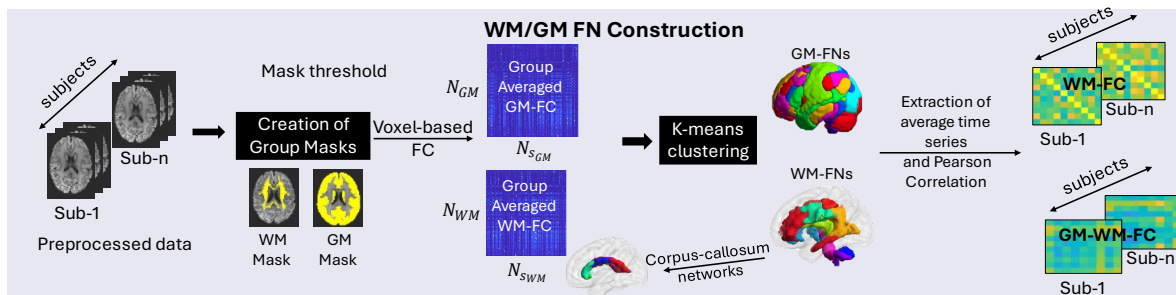


Figure 7-4 Flow Chart of WM/GM FN Construction

Once the FNs are constructed, the averaged time series corresponding to every FN is extracted and stored as .mat files.

White Matter FN Panel

WM-FN	GM-FN	Atlas-FC	Extract Time Series in frequency band
Individual WM Probability threshold: 0.6		Group WM threshold (%): 100	
K-means (WM) (K-Range): 2 ~ 22		Num CV folds: 4	
Sub-sampling the WM voxel FC matrix: Let toolbox decide			
☐ Corpus Callosum Networks		Create WM-FN	

Gray Matter FN Panel

WM-FN	GM-FN	Atlas-FC	Extract Time Series in frequency band
Individual GM Probability threshold: 0.6		Group GM threshold (%): 20	
K-means (GM) (K-Range): 2 ~ 22		Num CV folds: 4	
Sub-sampling the GM voxel FC matrix: Let toolbox decide			
		Create GM-FN	

7.2.1 Group Masks thresholds

The Individual WM/GM probability Threshold and the Group WM/GM threshold (%) are parameters to create the Group WM/GM Masks. These thresholds decide which voxels to consider in the Group WM/GM Masks.

All voxels for which at least Group WM/GM threshold (default 100% for WM and 20% for GM) of subjects have WM/GM probability greater than an Individual WM/GM probability Threshold (default 0.6) are included in the group GM/WM mask.

The highest value that the Individual WM/GM probability can take is 1. Higher values will lead to a smaller WM/GM Mask.

The highest value that the Group WM/GM threshold (%) can take is 100. Higher values will lead to a smaller WM/GM Mask.

7.2.2 K-means parameters

WhiFuN requires the user to specify a K-value range (default 2 to 22). For every K-value in the range, the average dice coefficient and the distortion value are computed to help the user find the optimal K-value.

If the user inputs the same value for the lower and the higher range (Eg. 6 ~ 6) then WhiFuN will skip the cross validation and directly create clusters with the value mentioned (Eg. 6).

A cross-validation approach is applied to determine the optimal number of clusters (K). The user can specify the number of cross validation (Num CV folds (default 4)).

Based on the memory available in RAM, WhiFuN can decide whether to use the complete voxel FC matrix or use a subsampled version of the matrix proposed by (Peer et al., 2017).

7.2.3 Corpus Callosum Networks

Under the WM-FN tab, there is a checkbox for Corpus Callosum networks. If this is checked, WhiFuN will divide the corpus callosum into sub regions that are maximally correlated to the created WM-FNs. The method proposed by (Wang et al., 2020) is used to create the Corpus callosum networks.

7.2.4 Create WM/GM FN

Once all the parameters are set, users can click on the Create WM-FN or GM-FN button.

- 1) WhiFuN will first create the Group WM/GM mask. The Group Masks will be stored in the <output_folder_path>/Analysis/Group_Masks as WMmask_allsubjs.nii and GMmask_allsubjs.nii for Wm and GM respectively.
- 2) Once the Masks are created, WhiFuN will compute the average voxel-level FC using the data from all the subjects that were selected with good data. Thereafter WhiFuN uses the cross validation approach proposed by (Peer et al., 2017) to find the optimal value of K for the K-Means clustering (this step will be skipped if the user inputs the same value for the lower and the higher range of K values).

- 3) After the cross-validation Grid search is complete for all the values of K, WhiFuN will plot a graph of Dice coefficients and Distortion values vs the K values.
- 4) WhiFuN will have a pop up asking the user what value of K should be chosen See Figure 7-5.
- 5) An optimal K-value has a high average dice coefficient and low distortion value. As one can observe from Figure 7-5, there are local peaks of Average Dice Coefficients. The K values corresponding to these peaks can be chosen as the optimal K value. These plots are saved as .png files in

<output folder>/Analysis/WM_FN/K_grid_search_dice_coefficient_WM.png

<output folder>/Analysis/GM_FN/K_grid_search_dice_coefficient_GM.png

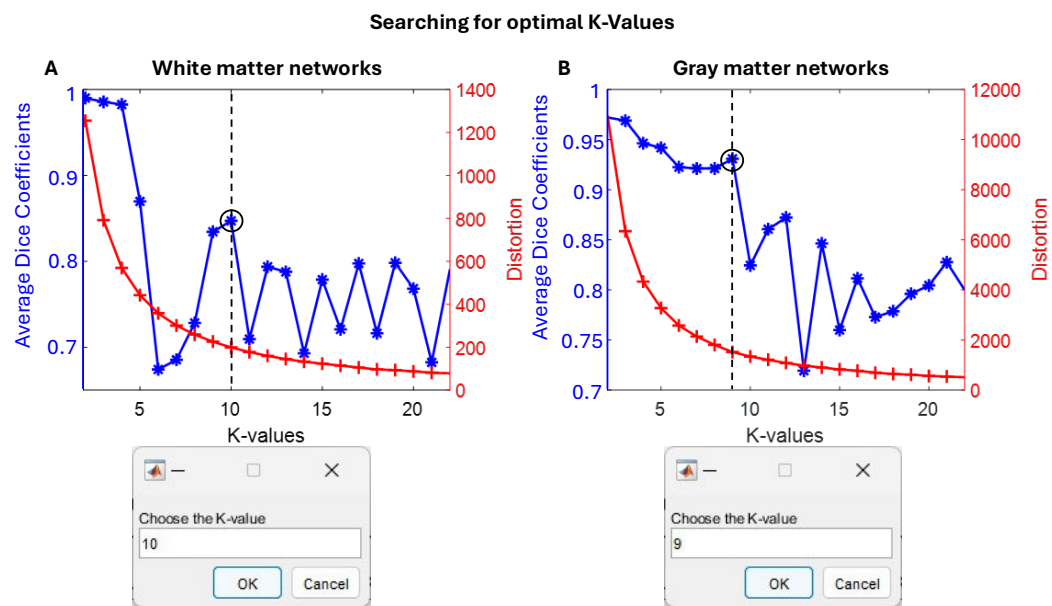


Figure 7-5: Finding the optimal number of FNs (K value). A: for WM-FNs and B: for GM-FNs. For this example K= 10 and 9 can be chosen as optimal K values for WM-FNs and GM-FNs respectively

- 6) Once the optimal K-value is selected, WhiFuN will compute the FNs and store the FN as a .nii file with name WM_clustering_K <value of K used>.nii for WM-FNs and GM_clustering_K<value of K used>.nii for GM-FNs.
- 7) The average BOLD time series of every WM and GM network for every subject are stored as a network_avgts_wm_K<value of K used>.mat and network_avgts_gm_K<value of K used>.mat file in

<output folder>/Analysis/WM_FN for WM

<output folder>/Analysis/GM_FN for GM

- 8) WhiFuN also stores 6 views of the WM-FNs in a folder WM_clustering_K<value of K used>_BrainNet_images in the WM_FN folder.

And the same for GM -FNs in the folder GM_clustering_K<value of K used>_BrainNet_images in GM_FN folder.

- 9) If the Corpus Callosum (CC) networks checkbox was checked, the CC-FNs will be stored in
<output folder>/Analysis/Corpus_Callosum_FN/CC_network_from_WM_K<value of K used for WM-FNs>.nii

The Mask used to identify the Corpus Callosum voxels is taken from Mori atlas (Mori et al., 2008) and is stored in the Corpus_Callosum_FN folder as

Corpus_callosum_atlas_MNI_<voxel dimension of MNI space>mm.nii

- 10) The average time series from every CC-FN is stored in the Corpus_Callosum_FN folder as network_avgts_cc_K<value of K used for WM-FNs>.mat

- 11) WhiFuN also stores the time series corresponding to every voxel in CC as cc_signals.nii.

7.3 Atlas based Functional Connectivity

If users have a predefined atlas and they wish to compute the FC matrix using the atlas, WhiFuN's Atlas-FC module can be used.

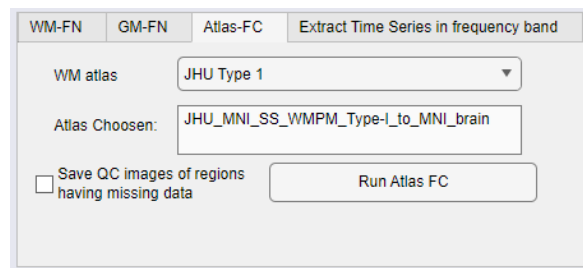


Figure 7-6: WhiFuN's Atlas FC module.

Here the user can choose the atlas. WhiFuN has already included JHU type 1, type 2 and type 3 atlases, but the user can also specify any other atlas (.nii file).

7.3.1 Quality Control for Atlas Based FC

Once the atlas is specified, the user can opt to save QC images for the subjects that have some NAN or missing values in regions of the atlas. An example of the QC image is shown in Figure 7-7. It shows the functional image for a particular subject (in this case Subject 0050952) and Atlas region 19 from the JHU type 2 atlas, with the contour of the atlas region overlapped on the Subject's functional image. It can be observed that this subject does not have data for few voxels that are in the region. Thus, the number of voxels used to get the average time series of this region for this subject will be less, which can be a problem.

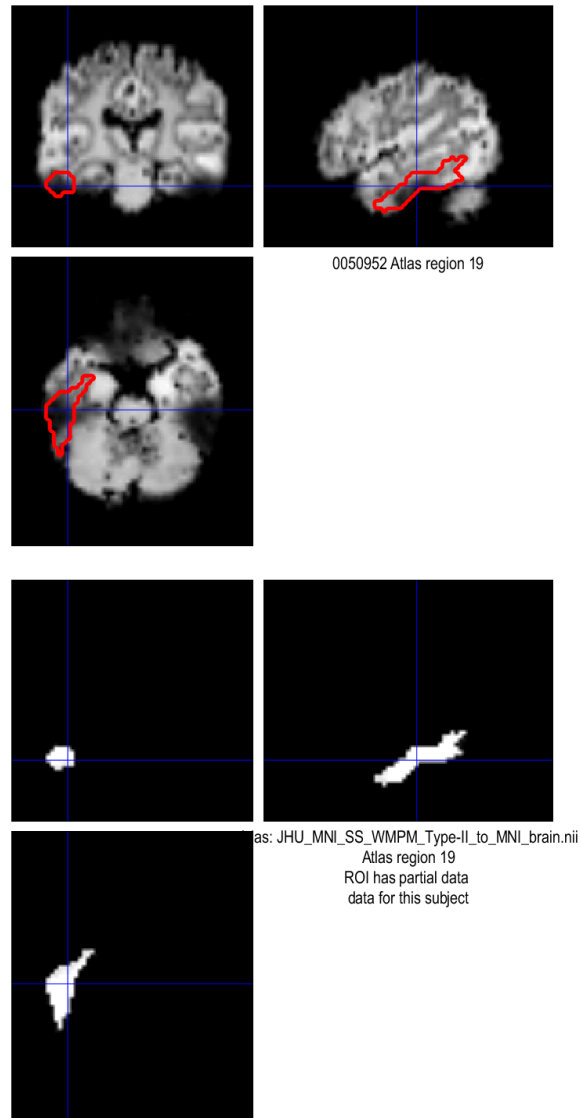


Figure 7-7: Quality Control for Atlas-FC

7.3.2 Run Atlas FC

Once the user Clicks on the Run Atlas FC button, WhiFuN will compute the average time series corresponding to every region in the atlas for every subject which can be used to compute the FC matrix using Pearson Correlation. (See *Figure 7-8*)

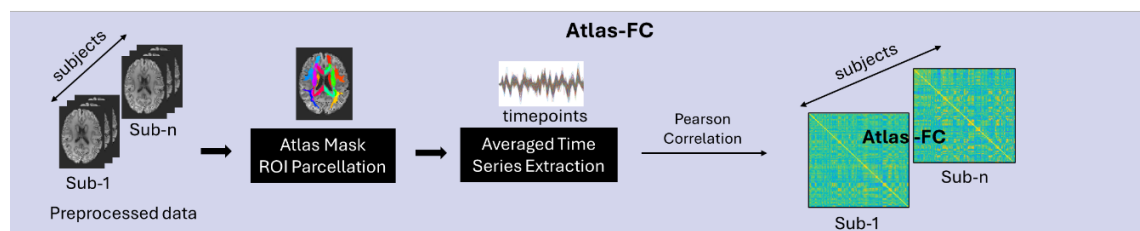


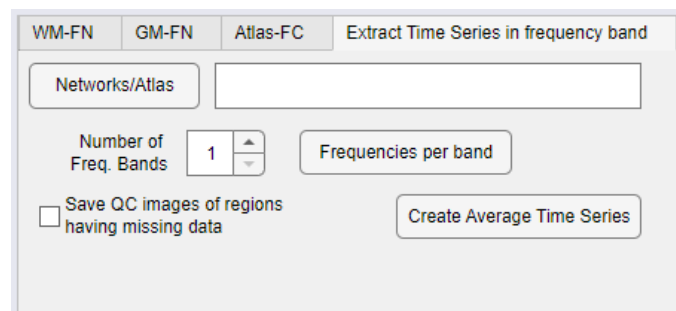
Figure 7-8: Atlas-FC computation

The average time series is stored as a .mat file with name <atlas_name>_reg_ts.mat in <output_folder>/Analysis/Atlas_FC folder.

If the dimensions of the specified Atlas is different from that of the functional images, WhiFuN will reslice the atlas and then use the resliced version of the atlas to compute the average timeseries. The atlas that is used to compute the average time series will also be stored in the Atlas_FC folder.

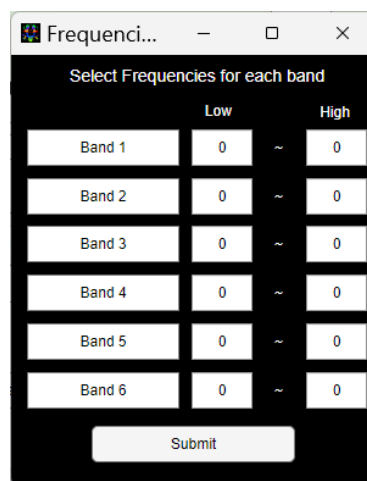
7.4 Extract Time series in frequency band

Users can extract the average time series in a particular frequency band, to study the frequency specific characteristics.



The screenshot shows a GUI window titled 'Extract Time Series in frequency band'. It has four tabs: 'WM-FN', 'GM-FN', 'Atlas-FC', and 'Extract Time Series in frequency band'. The 'Extract Time Series in frequency band' tab is active. Inside the window, there is a 'Networks/Atlas' button and a text input field. Below this, there is a 'Number of Freq. Bands' spinner set to '1' and a 'Frequencies per band' button. At the bottom left, there is a checkbox labeled 'Save QC images of regions having missing data'. At the bottom right, there is a 'Create Average Time Series' button.

Users can specify the atlas and the number of frequency bands. Then by clicking on the Frequencies per band button a new window pops up where the user can specify the different frequency bands and the names corresponding to these frequency bands and click submit. (See Figure 7-9)



The screenshot shows a window titled 'Frequencies for each band'. It has a title bar with a close button. The window contains a table with the following structure:

	Low		High
Band 1	0	~	0
Band 2	0	~	0
Band 3	0	~	0
Band 4	0	~	0
Band 5	0	~	0
Band 6	0	~	0

At the bottom of the window, there is a 'Submit' button.

Figure 7-9: Frequencies for each band GUI

QC plots as mentioned in Section 7.3.1 can also be created here by checking the Save QC images of regions having missing data checkbox.

7.4.1 Create Average time series

Once the user clicks the Create Average time series button

- 1) WhiFuN creates the frequency specific time series in the following Atlas folder
`<output_folder>/Analysis/Extract_timeseries/<atlas_name>`
- 2) The frequency specific time series are stored as
`<atlas_name>_<band_name>_reg_ts.mat`.
- 3) Additionally, the magnitude and phase response of the filter used is also stored in
`<output_folder>/Analysis/Extract_timeseries/<atlas_name>/Filters`
With name `<band_name>_filter_freq_response.png`.
- 4) The WM/GM and CC FNs created by WhiFuN (see Section 7.2) can also be used here to extract the frequency specific time series.

8. Analyze Functional Networks and Functional Connectivity

This module analyses the FNs and FC created during the Construct FNs and FC module by finding significant connections using statistical tests in FCs, checking how symmetric the FNs are across the two hemispheres or comparing the FNs computed by WhiFuN with any other predefined atlas to understand the network labels.

8.1 Compare FNs

The compare FNs module can be used to compare two sets of FNs. This function compares the FNs based on the Dice coefficient for two sets user specified FNs. This feature is helpful if the user wants to label the FNs by comparing them to a predefined atlas. We explain the working of this module by demonstrating how the WM-FNs generated by WhiFuN can be compared to the JHU-DTI81 atlas (Mori et al., 2008).

When the user clicks the Compare FN button, WhiFuN asks the user to select the first network. Users are expected to select the nifti file corresponding to the first network (See Figure 8-1). For instance, we select the WM-FN created by WhiFuN (See Section 7.2 for details) as the first network.

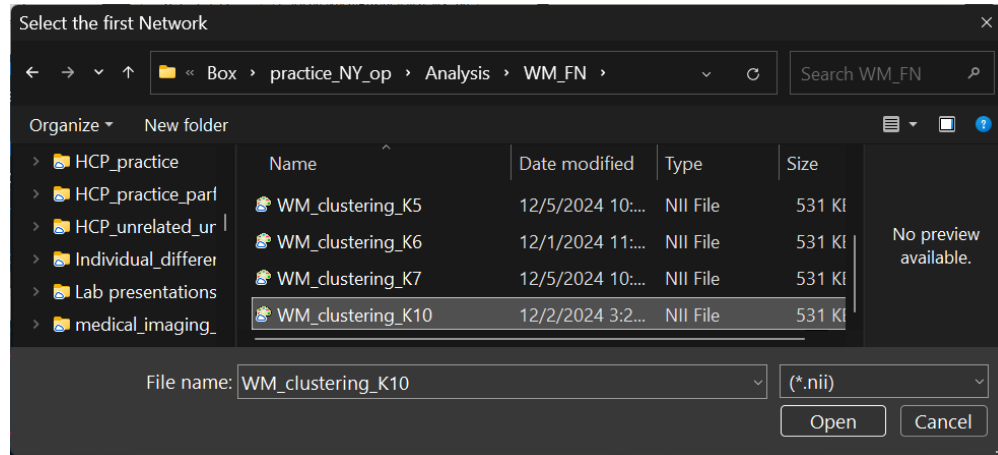


Figure 8-1: Select the First Network

Once the user clicks open, WhiFuN asks the user to select the second Network. We select the JHU-DTI-81 atlas (Mori et al., 2008).

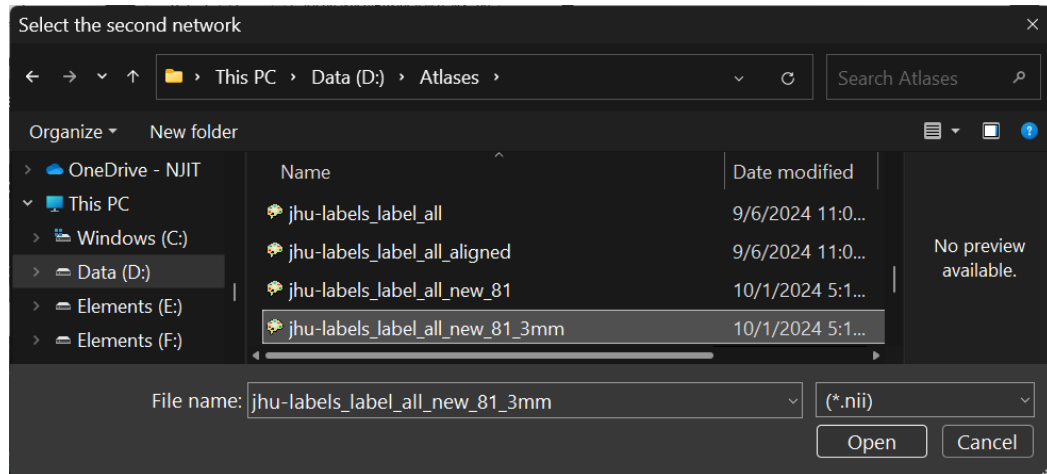


Figure 8-2: Select the Second Network

Once the user selects open, WhiFuN plots an $n_1 \times n_2$ matrix consisting of the Dice coefficient values calculated between every FN from the first set with that of the second set, where n_1 is the number of FNs in the first set and n_2 is the number of FNs in the second set.

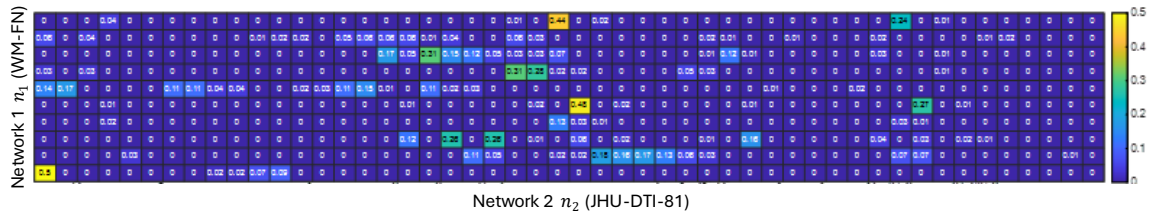


Figure 8-3 Result of Compare FN. Dice coefficients computed between every FN from the first set with that of the second set, where n_1 is the number of FNs in the first set and n_2 is the number of FNs in the second set

8.2 Check FN Symmetry

Using the Check FN Symmetry module, WhiFuN can calculate how symmetric the FNs are between the two hemispheres.

Once the user clicks the Check FN symmetry button, WhiFuN asks the user to select the network whose symmetry is to be computed (.nii file) (See Figure 8-4). For instance, we select the WM-FN created by WhiFuN (See Section 7.2 for details)

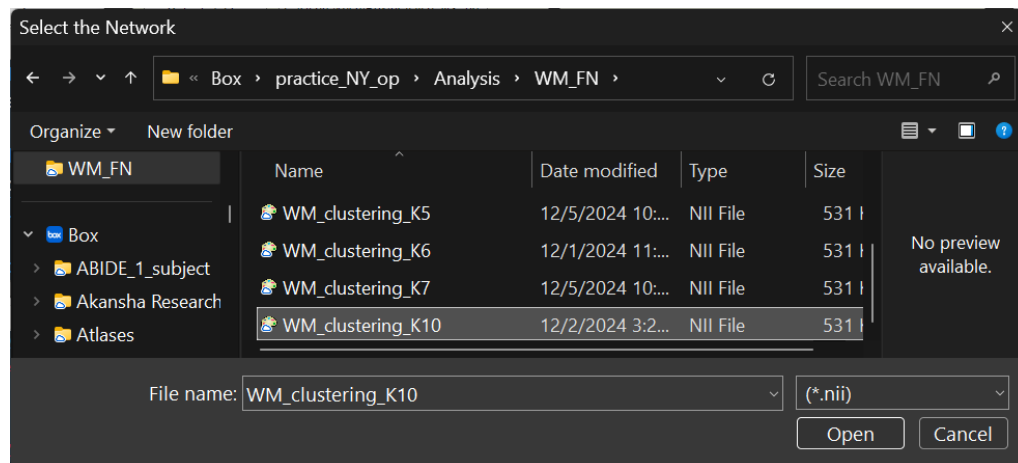


Figure 8-4: Select the network whose symmetry is to be computed.

To compute the symmetry, WhiFuN cuts the given network into two hemispheres (Left and right).

Once the user selects the network WhiFuN, WhiFuN plots an $n_1 \times n_1$ matrix consisting of the Dice coefficient values calculated between every network in the left hemisphere with that on the right. Where n_1 is the number of networks.

WhiFuN also calculates the total symmetry of the entire .nii file (Using the method proposed by (Peer et al., 2017) and puts it in the title of the resultant plot (See Figure 8-5)

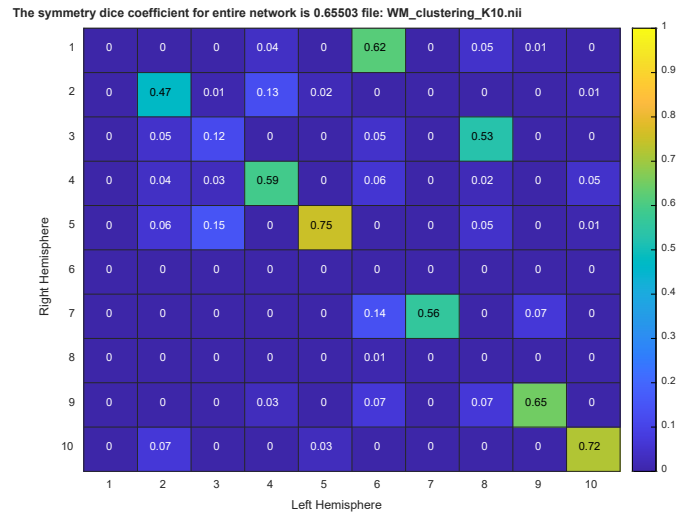


Figure 8-5: Result of Check FN symmetry. Dice coefficients computed between every network in the left hemisphere with that on the right

8.3 Statistics

WhiFuN can perform statistical tests (using MATLAB's regstats function) to find significant differences between groups of subjects or how well a particular FC can explain a behavioral score. To avoid false positives due to multiple comparisons, false rate discovery (FDR) is computed using mafdr (the Benjamin and Hochberg method) or Bonferroni correction computed by correcting the alpha value based on the number of tests that are performed are included in the module.

8.3.1 Upload CSV file

After the WM,GM or CC FNs, or Atlas-FC (See Section 7 for details) is created, users can click on the statistics button to open the WhiFuN statistics module (See *Figure 8-6*).

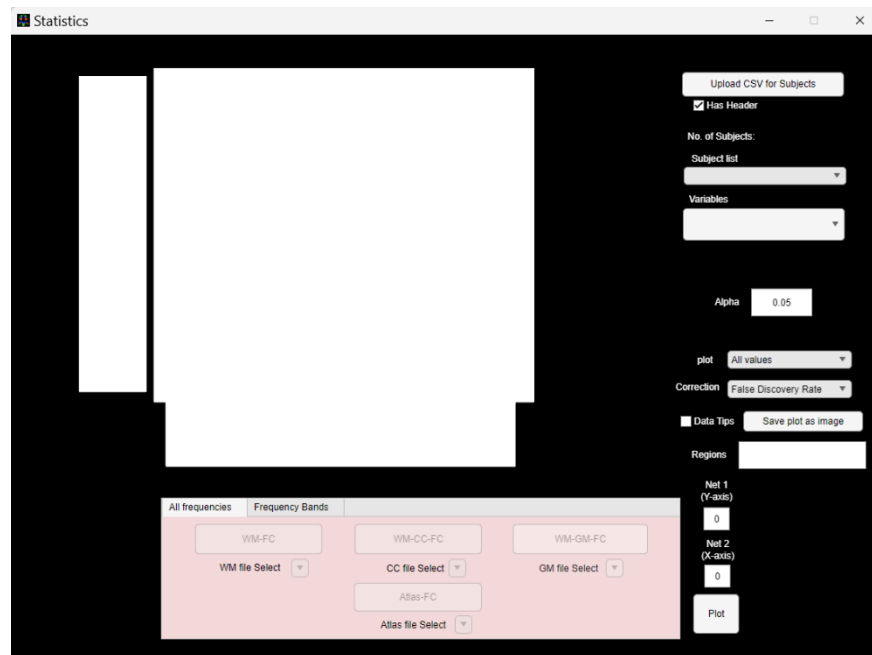


Figure 8-6 WhiFuN Statistics GUI

- 1) First the user will have to upload a CSV file. The first column of the CSV file should be Subject ID. Subject ID mentioned in the first column of the CSV should be the same as that of the Subject folder names.
- 2) The other columns of the CSV file can be any other metadata such as age, sex, or any other behavioral score. Ensure that except the Subject ID, all other columns should have numerical values. (For instance, for sex, Male can be represented by 0 and Female by 1).
- 3) Below we show an example of the CSV file for the practice dataset that has been used here. (Note that these are random sex and age values and not the actual).

Subject List	sex	age
0050952	0	22
0050953	1	25
0050954	0	26
0050955	1	28

- 4) Users can click on the Upload CSV for Subjects and choose the CSF file (See *Figure 8-7*).

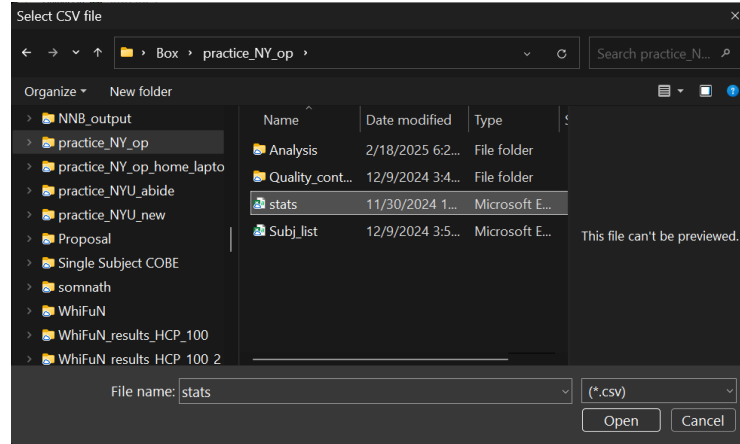


Figure 8-7: Select CSV file.

8.3.2 Statistics Module Features

Once the CSV file is uploaded the FC buttons (at the bottom of the GUI) corresponding to which the time series were stored will be enabled. If any of the buttons are not enabled that means WhiFuN did not find the time series corresponding to that particular FC creation.

- 1) Subject list drop down: This drop down shows the Subject IDs of the subject whose data is used in computation of the statistics.
- 2) Variables drop down: This drop down shows the variable that is being used to compute the statistical difference (T values) scores. The variables correspond to the columns of the CSV file that was uploaded except the 1st column (Subject ID).
- 3) Alpha: This is the significance alpha threshold. Default is 0.05.
- 4) Plot drop down: The user can choose to see the Statistics scores for every connection (All values) or only those that are significant (p-values less than Alpha).
- 5) Correction drop down: The users can choose the multiple comparisons corrections, the False rate discovery (FDR) (computed using MATLAB function mafdr (the Benjamin and Hochberg method) or Bonferroni correction (computing a new Alpha, by dividing it by the number of comparisons done).

The number of comparisons is computed by giving the c different p values obtained from every FC element. For WM-FC, if there are n_{WM} WM networks, $c = \frac{n_{wm} \times (n_{wm} - 1)}{2}$, for WM-CC-FC $c = n_{wm}^2$ and for WM-GM-FC $c = n_{wm} \times n_{gm}$, where n_{gm} is number of GM networks. For atlas-FC, with atlas having n_{ROI} number of ROIs, $c = \frac{n_{ROI} \times (n_{ROI} - 1)}{2}$.

- 6) Data Tips: When this is checked, the user can click on an element in the plot and see the corresponding T-value and p-Value.
- 7) Save plot as image: using this the user can save the plot as .png file for publications or posters.
- 8) Plot button: Users can see how different the FN connections are across the two groups (in this case sex). Since there are two groups, we see two bar plots, for n ($n < 5$) groups there will be n bar plots. If it is a continuous variable or if $n \geq 5$, it will show a scatter plot. User has to mention the network number from Y and X axis to get this plot.

- 9) Frequency Bands: Here the output from the Extract timeseries in specific frequency bands (see section 7.4) can be analysed.

See *Figure 8-8* to get a summary of all the features stated above. (We use the HCP 100 unrelated subjects dataset to demonstrate the working of statistics module below)

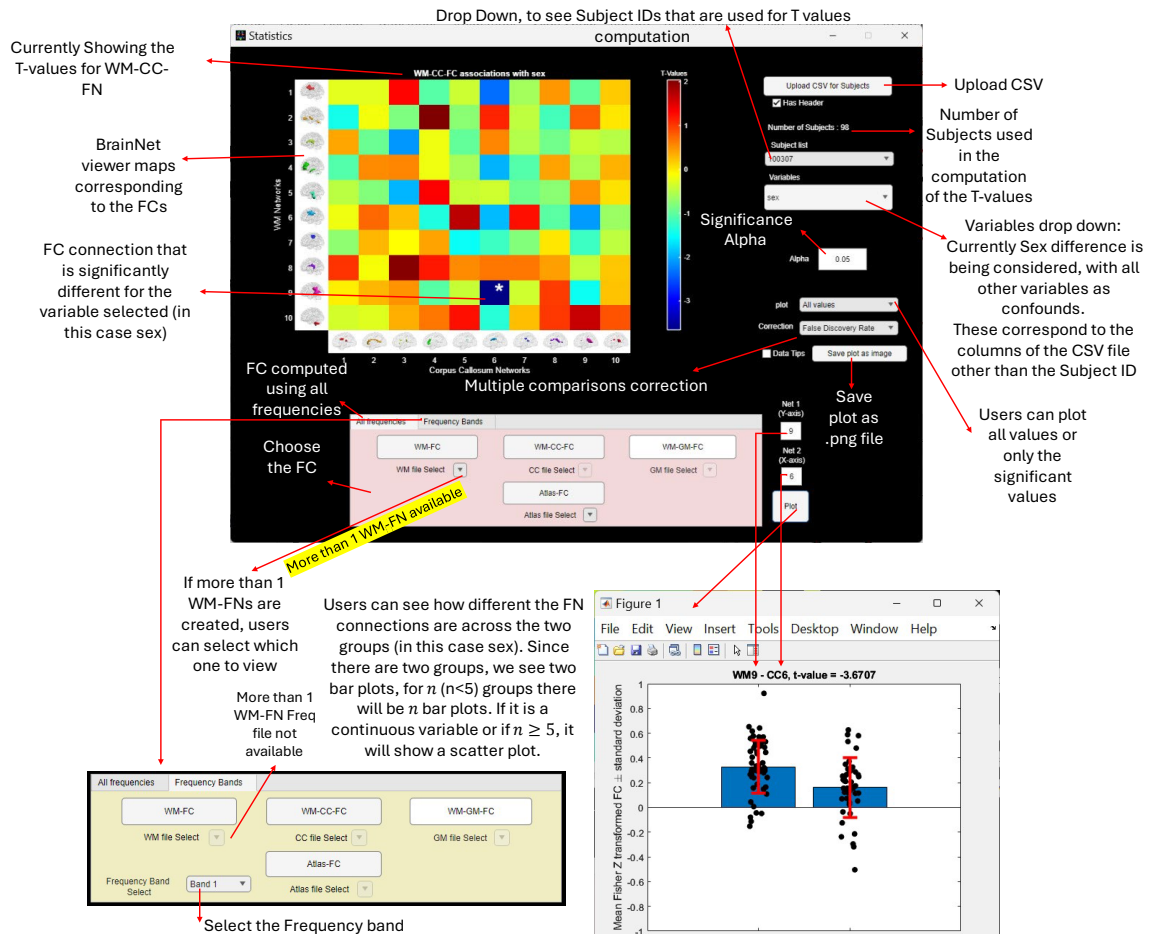


Figure 8-8: WhiFuN Statistics module Features

9. Visualizations

WM/GM FNs created can be visualized using the *Display FN* module, and the FC matrix can be visualized using the *Display FC* module.



Figure 9-1 WhiFuN Visualize module

9.1 Display FN

Once the functional networks are created using the details discussed in section 7.2, the networks can be viewed using the Display FN module.

As soon as the user clicks on the Display FN module, WhiFuN asks the user to select the clustering file (.nii file saved after WM/GM FNs are created). If the output folder is mentioned in the Setup, WhiFuN will directly open the WhiFuN Analysis folder. As an example, we show results from the HCP 100 unrelated subjects' dataset.

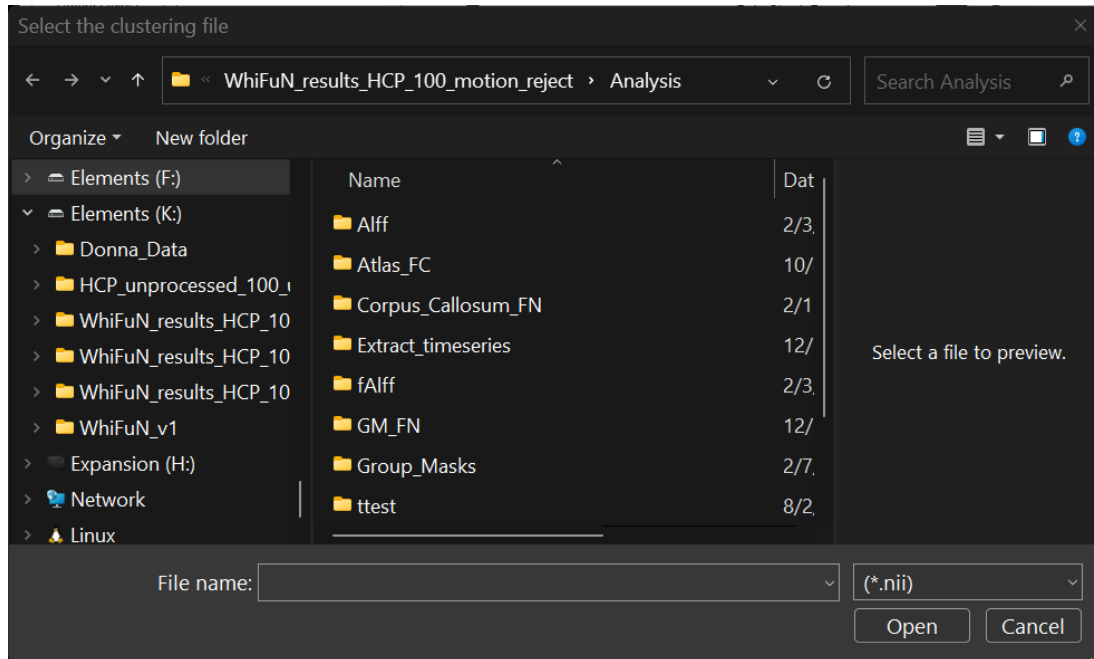


Figure 9-2 Select the FN file to visualize.

User can choose the desired FN file to visualize and click open.

WhiFuN can visualize the FN using the SPM display module or BrainNet viewer. WhiFuN will ask which toolbox the user would want to use to display the FN (see *Figure 9-3*).

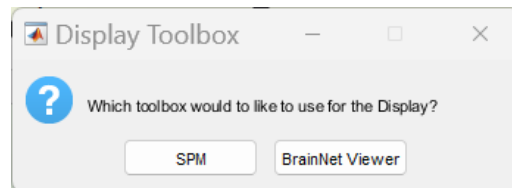


Figure 9-3 Which toolbox to use for Display FN.

9.1.1 SPM Display FN

If the user chooses SPM, A plot similar to *Figure 9-4* will be shown for the FN file selected. Users can move the crosshair and use any other SPM display feature here.

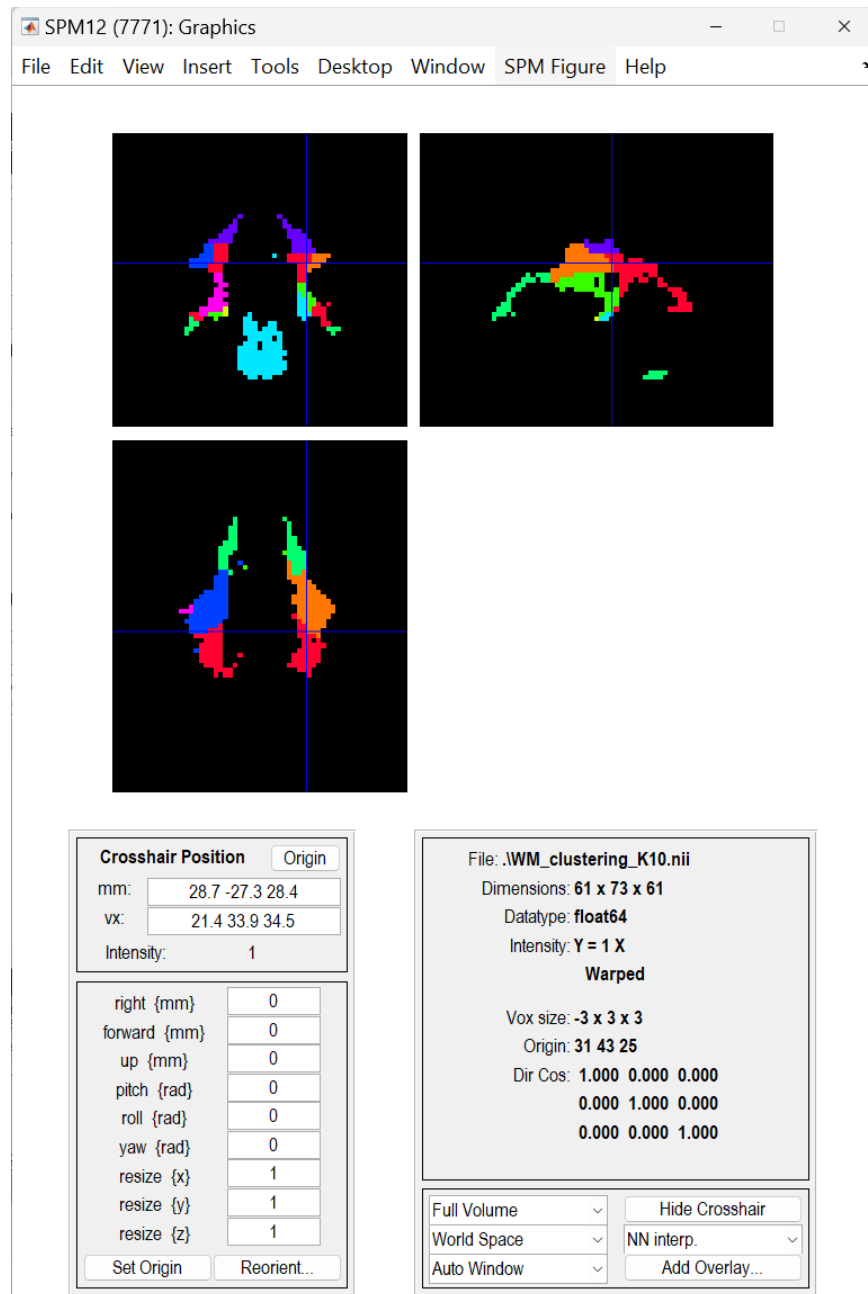


Figure 9-4 SPM display FN

9.1.2 BrainNet Viewer Display FN

If the user chooses BrainNet viewer, A plot similar to *Figure 9-5* will be shown for the FN file selected. Users can use any other features of the BrainNet viewer toolbox to see the plot as desired (For instance See *Figure 9-6*).

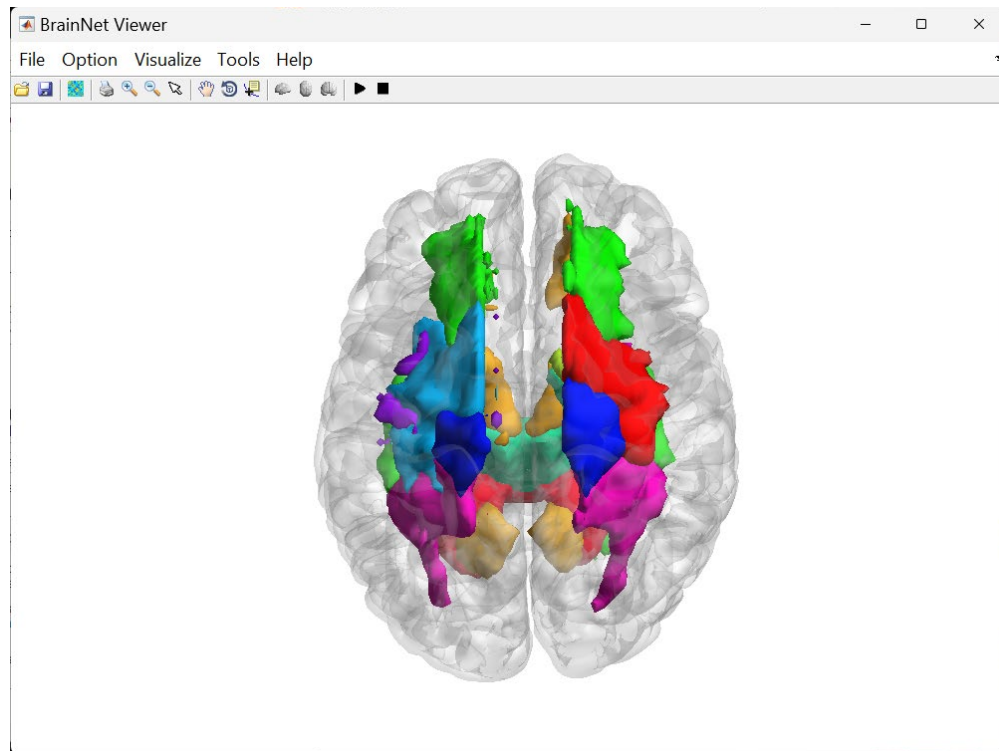


Figure 9-5 BrainNet viewer Display FN

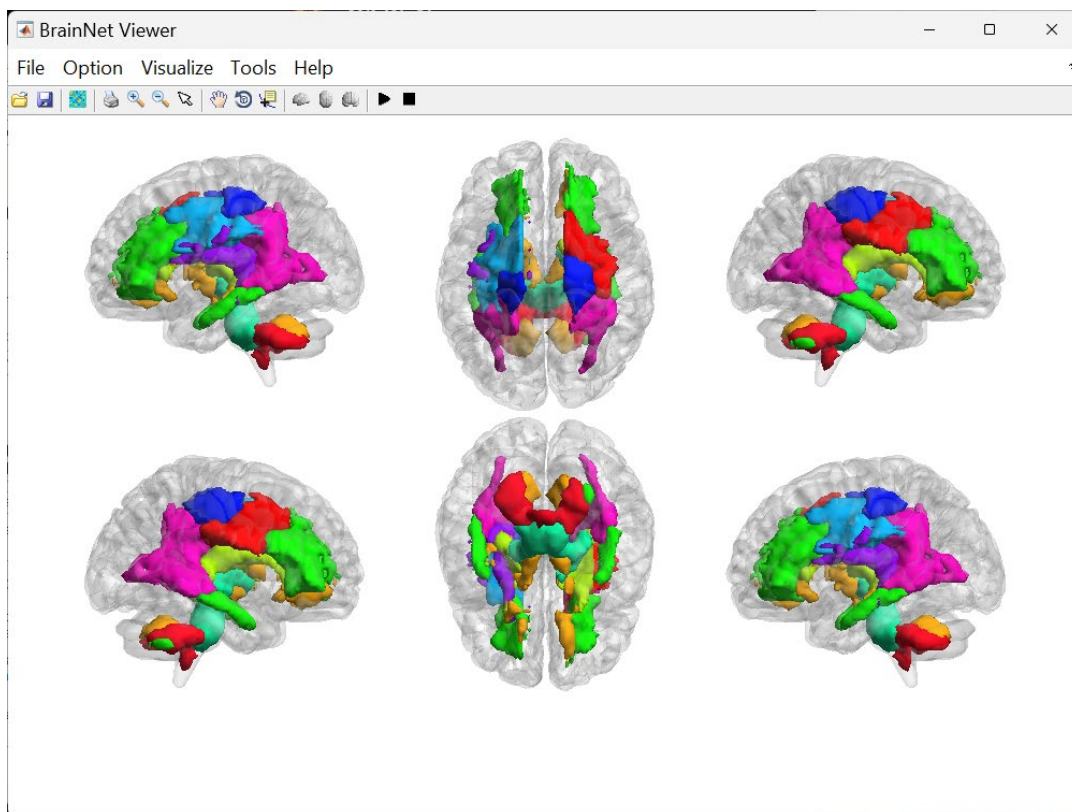


Figure 9-6: Click on option→ option→Layout→ Full view to see the above-mentioned plot.

9.1.2 Display FC

To visualize the FC matrices, the Display FC module can be used. Once the user clicks the Display FC button the Display FC GUI will open.

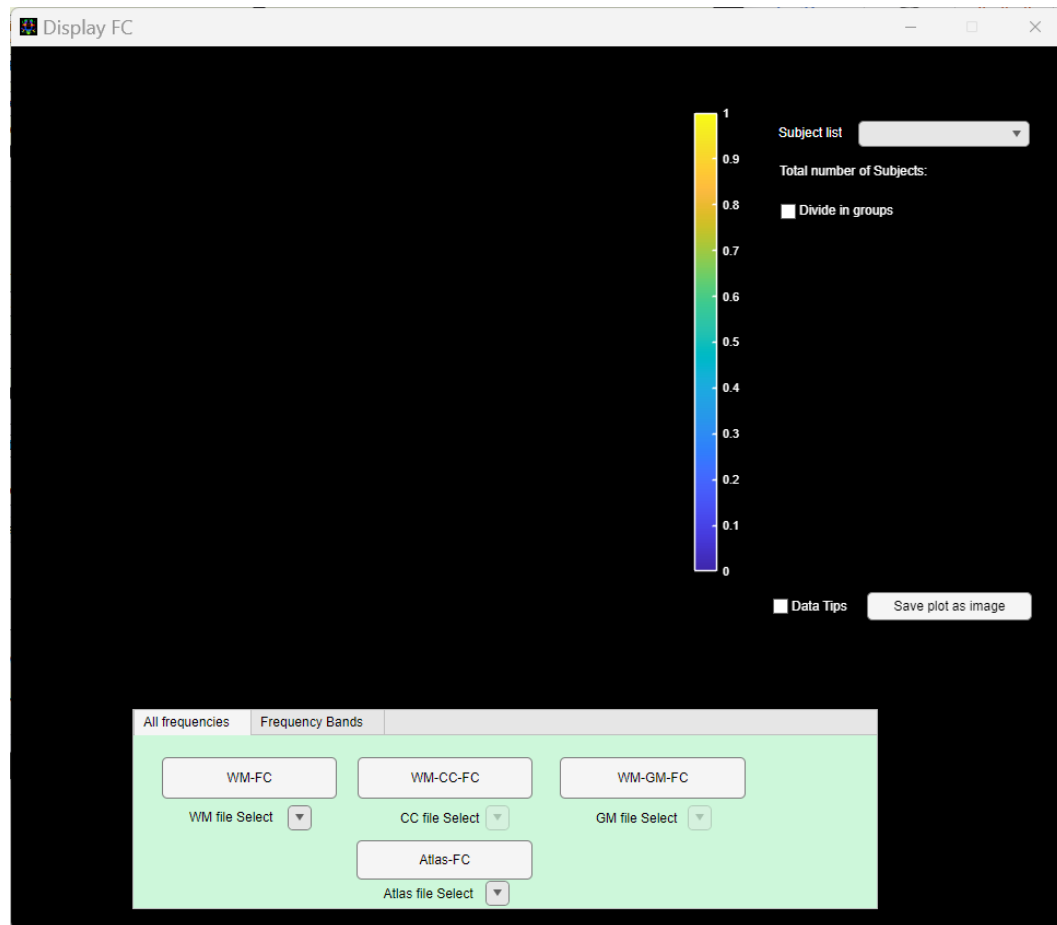


Figure 9-7: Display FC module.

Based on the average time series generated (see details in Section 7) the corresponding FC buttons (at the bottom of the GUI) will be enabled. If any of the buttons are not enabled that means WhiFuN did not find the time series corresponding to that particular FC.

Display FC Module Features

- 1) Subject List drop down: Users can use this drop down to choose to see the Mean, the Standard Deviation of the FC or the FC corresponding to individual Subjects.
- 2) Divide in Groups: Users can divide the subjects into two groups and then visualize the mean/STD of the FC from subjects in the group by checking this checkbox.

If Divide in Groups Checkbox is checked

- i) Upload CSV for Subject Groups: Upload a CSV file with two columns. The First column should be the subject ID (same as that in the Subject Folder), and the second column should have the groups. Only two groups are

allowed. So, the unique values that the 2nd column can have should be two. (For instance sex, can be M or F).

- ii) Name of group 1 or group 2: Users can put any name here, corresponding to Group 1 or Group 2 subjects.
 - iii) Group 1 or Group 2 subjects select: Alternatively the user can manually select subjects for group 1 and group 2.
- 3) Data Tips: When this is checked, the user can click on an element in the plot and see the corresponding FC value.
 - 4) Save plot as image: using this the user can save the plot as .png file for publications or posters.
 - 5) Frequency Bands: Here the output from the Extract timeseries in specific frequency bands (see section 7.4) can be visualized.

If this is checked, user can divide the subjects in 2 groups and visualize the mean/STD FC of the group



Figure 9-8 FC Display Module Features

10. Save/Load Parameters

Once WhiFuN is closed and a new session is open at a later date, the Load Parameters option can be used to fill all fields as it was when the first session was open.

After the user does the Run Data Check, a parameters.mat file is saved in the output folder. This file contains all the parameters used for preprocessing and setting up the data paths. Users can load this parameter and get all the fields populated.

Alternatively, users can also save the parameters at a different place or overwrite any other parameter.mat file using the Save parameters button.

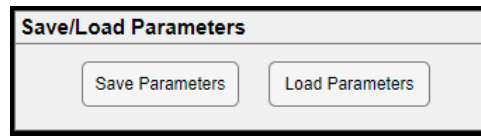


Figure 10-1: Save/Load Parameters

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